

Abstract

The rapid expansion of digital communication technologies, cloud computing infrastructures, Internet of Things (IoT) devices, and large-scale networked systems has significantly increased the exposure of modern computing environments to cybersecurity threats. As cyberattacks continue to evolve in complexity and frequency, conventional security mechanisms are often unable to provide adequate protection. Traditional intrusion detection systems typically rely on signature-based or rule-driven techniques, which are limited in their ability to detect unknown or emerging attacks. Furthermore, these approaches frequently struggle with scalability, adaptability, and real-time analysis when applied to large and dynamic network environments. Therefore, developing intelligent and adaptive cybersecurity solutions has become essential for effectively detecting and mitigating modern cyber threats.

Recent advancements in artificial intelligence and machine learning have shown promising potential for improving cybersecurity mechanisms. Machine learning–based intrusion detection models can automatically learn patterns from network traffic data and identify anomalies that may indicate malicious activities. However, many existing approaches still face several challenges, including high computational complexity, redundant feature representation, inefficient parameter tuning, and limited capability to adapt to evolving attack patterns. Additionally, models based on a single learning algorithm often lack robustness and may produce higher false alarm rates, which reduces their reliability in real-world cybersecurity applications.

To overcome these limitations, this research proposes an intelligent and optimized cybersecurity framework that integrates ensemble learning techniques, metaheuristic optimization algorithms, and reinforcement learning methods to enhance cyberattack detection and response mechanisms. The main objective of the proposed system is to achieve high detection accuracy, improved threat identification capability, and reduced computational overhead while ensuring adaptability to continuously changing network environments.

The proposed framework begins with a comprehensive data preprocessing stage designed to improve the quality and consistency of network traffic data. For experimental evaluation, a dataset consisting of 7,000 network traffic samples with 26 features is utilized. The preprocessing phase includes duplicate data removal, handling of missing values, and Min–Max normalization to standardize feature ranges and reduce noise in the dataset. These preprocessing steps improve the learning efficiency of the models and enhance the reliability of the detection system.

After preprocessing, an advanced feature extraction process is performed to capture important characteristics of network behavior. The proposed approach extracts multiple types of features, including statistical attributes, temporal characteristics, network traffic indicators, behavioral patterns, and entropy-based measurements. This comprehensive feature representation enables the detection framework to effectively differentiate between normal network activities and malicious attack patterns.

To further enhance feature quality and reduce dimensionality, hybrid feature selection and optimization techniques are employed. The Zebra Optimization Algorithm (ZOA) and Pelican Optimization Algorithm (POA) are applied to identify the most informative features and eliminate redundant attributes. In addition, the Prairie Dog Optimization (PDO) algorithm is used for optimizing the hyperparameters of the ensemble machine learning classifiers. These optimization techniques improve model performance while reducing computational complexity.

Using the optimized feature set, an ensemble learning–based intrusion detection model is developed to strengthen classification capability and improve detection reliability. Ensemble learning combines multiple machine learning models to enhance generalization performance and increase detection robustness compared to individual classifiers. Experimental results indicate that the proposed ensemble model achieves an accuracy of 99.7%, precision of 99.4%, and recall of 98.4%, demonstrating its effectiveness in accurately identifying and classifying cyberattack activities while maintaining a low false positive rate.

To further enhance the adaptability and intelligence of the system, a reinforcement learning module is incorporated into the framework for dynamic threat detection and mitigation. The reinforcement learning architecture integrates Double Deep Q-Network (DDQN) for anomaly detection, Multi-Agent Deep Deterministic Policy Gradient (MADDPG) for cooperative multi-agent analysis of threats, and Proximal Policy Optimization (PPO) for stable and adaptive policy learning. Additionally, feature optimization within the reinforcement learning module is performed using a hybrid optimization strategy that combines the Snake Optimizer (SO) and Harris Hawks Optimization (HHO) algorithms.

Extensive experimental evaluation demonstrates that the proposed reinforcement learning–enhanced cybersecurity framework significantly outperforms existing intrusion detection approaches. Compared with conventional reinforcement learning models, the proposed system achieves a 4.8% improvement in detection accuracy, a 6.3% increase in detection rate, and an 18.5% reduction in

computational overhead, making it highly suitable for real-time cybersecurity applications in large-scale network environments.

In summary, this research introduces an advanced cybersecurity framework that integrates ensemble machine learning, hybrid optimization algorithms, and reinforcement learning techniques to address the challenges associated with modern cyberattack detection. By improving detection performance, reducing computational complexity, and enabling adaptive decision-making capabilities, the proposed system provides an effective and scalable solution for protecting modern network infrastructures against evolving cyber threats.

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Abbreviations

AI	Artificial Intelligence
APT	Advanced Persistent Threats
ANN	Artificial Neural Network
CAN	Controller Area Network
CPS	Cyber Physical Systems
CNN	Convolutional Neural Networks
CTI	Cyber Threat Intelligence
DDoS	Distributed Denial of Service
DCGAN	Deep Convolutional Generative Adversarial Networks
DL	Deep Learning
DQN-MAFS	deep Q-network multi-agent feature selection
DNS	Domain Name System
DNN	Deep Neural Networks
DT	Decision Tree
EDVC	Ensemble Deep Voting Classifier
FPR	False Positive Rate
FSS	Feature Self-Selection
FNR	False Negative Rate
FTC	Fault-Tolerant Control
FTP	File Transfers Protocol
GBDT	Gradient Boosted Decision Trees
GCN	Graph Convolutional Network
GRU	Gated Recurrent Units
GMADRLD	Grouped Multi-Agent Deep RL Defense
HHO	Harris Hawks Optimization
HZPOA	Hybrid Zebra Pelican Optimization Algorithm
KNN	K-Nearest Neighbours
IAM	Identity and Access Management

IDS	Intrusion Detection Systems
IoT	Internet of Things
IPS	Intrusion Prevention Systems
LSTM	Long Short-Term Memory
LR	Logistic Regression
MADDPG	Multi-Agent Deep Deterministic Policy Gradient
MAFSIDS	Multi-Agent Feature Selection Intrusion Detection System
ML	Machine Learning
MITM	Man-in-the-Middle
MRMSVM	Mud Ring-aided Multilayer Support Vector Machine
PCA	Principal Component Analysis
PDO	Prairie Dog Optimization
POA	Pelican Optimization Algorithm
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RF	Random Forest
RIDS	Real Time Intrusion Detection System
RL	Reinforcement Learning
ROC-AUC	Receiver Operating Characteristic - Area Under the Curve
RNN	Recurrent Neural Networks
SAGI	Shapley Additive Global Importance
SAC	Soft Actor-Critic
SM	Slime Mold
SN	Snake Optimizer
SVM	Support Vector Machines
WSS	Weighted Score Selector
ZOA	Zebra Optimization Algorithm

Chapter1: Introduction

Anomaly detection is a critical process in cybersecurity, as it enables the identification of unusual activities that may indicate malicious behavior within networked systems. Conventional security solutions such as intrusion detection and prevention systems and firewalls provide basic protection; however, their dependence on predefined rules and signatures limits their ability to detect sophisticated and emerging cyber threats [1]. Contemporary cyberattacks, often initiated by organized groups or state-sponsored entities, target system vulnerabilities to disrupt operations, steal sensitive data, and compromise critical infrastructure. As the scale and complexity of cyber threats continue to grow, there is an increasing demand for intelligent security mechanisms that can adapt to changing environments and detect previously unseen attacks [2].

The rapid growth of interconnected technologies, including IoT devices, artificial intelligence, and data-driven systems, has significantly expanded modern network environments [3]. Cyber-Physical Systems (CPS), which integrate computational intelligence with physical processes and operate in real time, face heightened security risks due to their reliance on continuous network communication [4]. Attacks on such systems can result in severe consequences, including power outages, industrial failures, transportation disruptions, and large-scale information leakage. Additionally, the widespread deployment of low-cost connected devices with limited security features has increased vulnerability to advanced threats such as zero-day exploits and persistent attacks [5].

Machine learning has become an essential tool for strengthening cybersecurity by enabling systems to learn complex attack patterns directly from data [6]. Reinforcement Learning (RL) offers a dynamic approach by allowing an agent to learn optimal defense strategies through interaction with its environment. By utilizing a reward-based feedback mechanism defined by states, actions, and outcomes, an RL agent incrementally improves its decision-making capability [7]. This learning paradigm is especially effective in adversarial and real-time scenarios, where labeled data is limited and attack behaviors continually evolve.

Multi-Agent Reinforcement Learning (MARL) further enhances this approach by deploying multiple autonomous agents that collaborate within a shared environment. In distributed network infrastructures, MARL enables agents to perform specialized security functions such as traffic analysis, anomaly detection, and threat mitigation, thereby improving scalability and

overall detection accuracy [8]. Coordinated learning among agents reduces false alarms and strengthens resilience against complex and distributed attack patterns. Advanced deep reinforcement learning methods such as Double Deep Q-Networks (DDQN), Proximal Policy Optimization (PPO), and Multi-Agent Deep Deterministic Policy Gradient (MADDPG) contribute to improved stability, adaptability, and cooperative decision-making in dynamic cybersecurity settings [9].

In response to these challenges, this research presents a Multi-Agent Reinforcement Learning based framework for cyberattack detection and mitigation. The proposed approach integrates effective data preprocessing, optimized feature selection, and ensemble learning strategies to enhance detection performance while reducing computational overhead and false positive rates. A Centralized Training and Decentralized Execution (CTDE) mechanism is employed to enable collaborative learning during training and autonomous operation during deployment. By combining reinforcement learning with traditional machine learning classifiers, the framework aims to deliver a scalable, adaptive, and robust solution for securing modern network infrastructures against evolving cyber threats [10].

Chapter2 : Literature Survey

This chapter presents a comprehensive review of existing research related to cyberattack detection in modern network environments. The review primarily focuses on machine learning based intrusion detection systems, feature optimization techniques and ensemble learning methods used to enhance cybersecurity frameworks. A wide range of approaches has been explored to enhance the efficiency, scalability, and adaptability of cybersecurity threat detection systems. These approaches are generally classified into Machine learning and Reinforcement learning based.

2.1 Review of Machine learning based threat detection techniques

A brand-new ensemble-based ML method for intrusion recognition was created via Alamgir et al. [11] and is demonstrated. The performance of the suggested method will be assessed using a variety of ensemble methods and many publicly available datasets, including the Gradient XGBoost, Bagging, RF, AdaBoost, gradient booster, and Simply Stacking. Principal component analysis (PCA), shared data, and correlation analysis are used to choose the most pertinent features for intrusion detection.

Shereen et al. [12] introduce a Weighted Score Selector (WSS), a lightweight ensemble-based ML technique for identifying cyberattacks in WSNs. A combination of supervised ML classifiers is used to achieve the suggested method, whereby the best classifier is continuously promoted to improve detection performance as soon as possible. The performance of the suggested method was contrasted with three traditional ensemble methods: stacking-based, bagging-based, and boosting-based.

In our tests, Rasheed et al. [13] used deep learning (DL) models to identify novel hazards that had never been added to the model before. The bias concerns related to the detection of unknown threats were also examined in this work. This paper offers a realistic intrusion detection system (IDS) method that tests unknown attack scenarios by training an ensemble of DL classifiers on four reference IDS datasets. The suggested architecture uses classifiers that have been reported to process and yield decent results for sequential information to create a DL ensemble.

Alamgir et al. [14] develop a novel method for the enduring botnet issue by combining a strong collection of methods from ML with a Hybrid selection of features approaches, such as PCA,

complementary information, and categorical analysis. It improves the ensemble learners' detection capabilities by uniquely combining these feature selection approaches to improve the data space. As the preferred ensemble technique, Extra Trees performs admirably, achieving a nearly flawless accuracy in botnet detection across a variety of datasets.

The method routinely achieves more than 99 percent true-positive rates and an incredible FPR, setting a new standard for dependability in botnet detection, as demonstrated by detailed comparison assessments. This investigation represents a revolutionary advancement in cybersecurity, giving previously unheard-of levels of accuracy and resistance to botnet intrusions, as well as a crucial model for creating next-generation security systems.

The suggested genetic ensemble classifier, Muhammad Ali et al. [15] employs a mutual data analysis method using four strong ML ensemble procedures: the RF-based Bagging algorithm, the Gradient Boosting Classifier, the Bagging Classifier, and the Voting Classifier. A network dataset is employed to develop and test a model for every method. The graphs of rates of performance have been examined to evaluate both algorithms' performance regarding their capacity to predict the occurrence of anomalies.

Abdussalam et al. [16] propose an ensemble continuous ML approach to improve DDoS mitigation and detection. This method adapts the model with anticipated assault patterns using online learning. SDN simulation is used to train and assess the model (Mininet and Ryu). By overcoming traditional constraints, its dynamic choice of features improves accuracy across a variety of DDoS assault types. Results from the experiment show an astounding 99.2% detection rate.

Furqan et al. [17] propose an ensemble deep voting classifier (EDVC) to identify network threats accurately. The suggested method uses the majority voting criteria and uses recurrent neural networks (RNN), gated recurrent units (GRU), and long short-term memory (LSTM). According to the experiments conducted using the NSL-KDD dataset, EDVC outperforms the existing models with an accuracy score of 0.996.

A new method for detecting DDoS attacks is being introduced by Shravan et al [18]. Three different CNN systems are combined in the suggested method: SA-enabled CNN using RF, SA-enabled CNN using LSTM, then SA-enabled CNN using XGBoost. While self-attention techniques improve feature integration and significance, each model extracts information at several scales, by ensuring that both noticeable and subtle indicators are included in the final

categorization. The suggested technique detects DDoS attacks with 98.71% specificity and 98.69% reliability.

An effective automated IDS was created by Anil et al. [19] utilizing ML methodology. Min-max normalization and null value handling are employed to preprocess the collected data. Modified Singular Value Decomposition is used to extract features (M-SvD). The Opposition-based Northern Goshawk Optimization algorithm (ONgO) optimizes the extracted characteristics. A hybrid ML technique known as the mud ring-aided multilayer support vector machine (M-MultiSVM) categorizes the various attack types. As a result, network intrusion can be effectively detected by the suggested M-MultiSVM model.

For identifying and mitigating network intrusions, Mamatha et al. [20] developed a potent IDS based on DL. Deep Convolutional Generative Adversarial Networks (DCGAN) is a data augmentation technique that addresses the issue of data imbalance. After that, a CenterNet-based method extracts the characteristics from the input data. ResNet152V2 by Slime Mold (SM) technique-based DL is used to categorize the attacks. The suggested defense module is turned on as soon as the network intrusion is discovered to promptly restore normal network communication. The existing techniques and their advantages and disadvantages are described in Table 1.

Table 2.1: Summary of the existing ML models with their advantages and disadvantages

Author(s)	Technique	Advantage	Disadvantage
Alamgir et al. [11]	Ensemble methods	High accuracy using PCA, shared data, and correlation analysis for feature selection.	Performance is dependent on feature selection; the computational cost of ensemble methods may be high for large datasets.
Shereen et al. [12]	WSS	Lightweight ensemble method with adaptive classifier promotion for enhanced detection performance.	May struggle with highly imbalanced datasets or unknown attacks due to reliance on supervised classifiers.
Rasheed et al. [13]	DL ensemble for IDS	Effective detection of unknown threats and reduced bias in sequential information processing.	High computational cost due to DL, requires extensive training data for generalization.

Alamgir et al. [14]	Hybrid feature selection with Extra Trees	Excellent performance in botnet detection with over 99% TPR and 0% FPR, robust against varied datasets.	Limited scalability to real-time systems and reliance on preprocessed datasets may hinder adaptability.
Muhammad Ali et al. [15]	Robust Genetic Ensemble Classifier	A comprehensive evaluation of ensemble algorithms for anomaly detection, a flexible approach to combining classifiers.	Potential redundancy in using similar classifiers, may not address novel attack patterns effectively.
Abdussalam et al. [16]	Continuous online learning ensemble model	Adaptive to new DDoS patterns; achieves 99.2% detection accuracy, uses SDN simulation for realistic testing.	Performance dependent on the accuracy of simulated training environments, computationally intensive online learning.
Furqan et al. [17]	EDVC	Achieves state-of-the-art accuracy (0.996) for network threat finding by means of the NSL-KDD dataset.	Limited generalizability to other datasets and dependency on sequential data processing can increase latency.
Shravan et al. [18]	SA-enabled CNN with RF, LSTM, XGBoost	Effective feature integration for DDoS detection with 98.71% specificity and 98.69% reliability.	Computational overhead due to multiple CNN architectures, may require fine-tuning for different datasets or scenarios.
Anil et al. [19]	M-SvD with ONgO and M-MultiSVM	Optimized feature extraction and hybrid ML model achieve effective detection of network intrusions.	Complexity of optimization techniques (ONgO), potential difficulty in scaling to large datasets or real-time scenarios.
Mamatha et al. [20]	DCGAN, CenterNet, ResNet152V2 with SMA	Addresses data imbalance effectively, robust classification, and prompt threat mitigation via the defense module.	High computational requirements for DCGAN and ResNet152V2, and dependency on specific architectures may limit flexibility.

2.2 Review of Reinforcement learning based threat detection techniques

Thajeel, et al. in (2023) [20] presented a feature drift-aware approach for detecting Cross-Site Scripting (XSS) attacks. The suggested approach involves dynamic feature selection using a deep Q-network multi-agent feature selection (DQN-MAFS) framework. Here, an individual agent is linked with one feature and is accountable for choosing or rejecting its feature. Also,

the developed framework offers a sub-model, namely, a fair agent reward distribution based on dynamic feature selection for distributing rewards over several agents. Thus, the presented framework supports real-time dynamic adjustments of embedded knowledge as new data arrives.

Ren et al. in (2023) [21] developed a Multi-Agent Feature Selection Intrusion Detection System (MAFSIDS) based on DQL. The designed framework comprises a feature self-selection (FSS) mechanism and a DRL attack detection module. The FSS algorithm leverages a MARL framework that redefines the problem of feature selection by converting conventional $2N$ feature selection space into N agent representations. This lessens model complexity and enhances the search during feature selection. Then, a Graph Convolutional Network (GCN) is developed to extract deep features from the data, thereby ensuring accurate feature representation.

Tellache et al. in (2024) [22] suggested a new MARL framework that enables robust, automatic, and efficient detection of intrusions. Here, the efficiency of the presented model is enhanced by improving the DQN algorithm with the implementation of the mean square loss function and cost-sensitive learning approaches. Thus, the designed resilient architecture easily accommodates new attacks and adapts to changing attack patterns.

Han et al. in (2022) [23] developed an intrusion detection hyperparameter control system (IDHCS) for training and controlling a deep neural network (DNN)-based feature extractor framework and a PPO-based k -means clustering framework. At first, the designed framework effectively controls the feature extractor unit for capturing the most important features and detects intrusions with the help of k -means clustering. Then, the developed system is optimized using PPO-based RL to improve the performance.

Rajae et al. in (2023) [24] introduced a novel Multi-Agent Distributed Deep Learning (MADDL) approach to tackle cyber-attacks in distance relays. With the application of graph theory, the protection system with numerous distance relays is mapped to a multi-agent distributed system. Here, the distance relays are deliberated as agents or nodes of the multi-agent system in which an individual agent is connected to its nearby agents to exchange current and voltage details. Then, a DNN-based cyber-attack detection framework is used for detecting attacks using local voltage, current, and received data.

Yu et al. in (2023) [25] presented a deep reinforcement learning-based intrusion detection system (IDS) for IIoT environments. A DQN framework is intended to solve an open-set

recognition issue that is modeled as a discrete-time Markov decision process. Here, the problem is divided into 2 subproblems such as known traffic fine-grained classification problem and unknown attacks recognition problem. In this context, DQN is employed to solve known traffic fine-grained classification problems, and a reconstruction error is used to identify unknown attacks.

Chen et al. in (2025) [26] used the MADRL method to solve the issues of high uncertainty and changing aspects of attacks. Here, all the defenders are divided into cooperative groups, and defenders within each group are allowed to optimize the defense strategy through information sharing. In this context, a novel grouped multi-agent deep RL defense (GMADRLD) approach is used to mitigate state space explosion issues, thereby attaining good defense effects.

Zhang et al. in (2023) [27] suggested a DDQN framework to detect anomalies in the traffic. Here, an agent was trained to identify attack patterns that do not match legitimate traffic on the end side. The presented framework identifies known and unknown attacks by engaging directly with a simulation environment. Here, the usage of a limited amount of labeled attack data is concurrently optimized for better understanding and helps in the detection of unlabelled attacks.

Mohamed et al. in (2023) [28] presented a DRL model for detecting intrusions by combining SARSA-based RL algorithm with a DNN. The offered model accurately detects modern and complex attacks in the network environment with high detection accuracy.

Liu et al. in (2022) [29] applied a DRL algorithm to design an FDI attack detector. Initially, the problem of discontinuous attack detection is modelled as a partially observable Markov decision process (POMDP). Here, a neural network is used to explore this problem.

Tareq et al. in (2024) [30] used a DRL framework to simulate harmful cyber-attacks and improve cyber-security. The designed framework utilizes an agent-based model for continuous learning and adaptation within dynamic network surroundings. Here, the agent estimates an optimal course of action depending on state and corresponding rewards received for making decisions. The existing techniques and their advantages and disadvantages are described in Table 2.

Table 2.2: Summary of the existing RL models with their advantages and disadvantages

Author & Year	Methods	Advantages	Disadvantages
Thajeel, et al. (2023) [20]	DQN-MAFS with dynamic feature selection.	Ensures real-time adjustments to fresh data.	Suffers from high computational complexity.

Ren et al. (2023) [21]	MAFSIDS using DQL and GCN.	The FSS mechanism decreases search space complexity, and GCN increases feature extraction.	Performance hinges on accurate feature representation.
Tellache et al. (2024) [22]	MARL with an enhanced DQN.	Supports cost-sensitive learning and guarantees efficient intrusion detection.	Face challenges in real-time applications with changing attack patterns.
Han et al. (2022) [23]	IDHCS uses PPO and k-means clustering.	Provides optimized feature extraction and intrusion detection.	Requires widespread computational resources.
Rajae et al. (2023) [24]	MADDL using DNN and Graph Theory.	Supports efficient cyber-attack detection.	High communication overhead creates latency.
Yu et al. (2023) [25]	Deep RL-based Intrusion Detection System using DQN.	Efficiently handles open-set recognition for recognizing unknown attacks.	Reconstruction errors could misclassify complex attacks.
Chen et al. (2025) [26]	GMADRLD with cooperative learning.	Effectively mitigates the state space explosion.	Suffers from high complexity and requires large computational resources.
Zhang et al. (2023) [27]	DDQN for detecting anomalies.	Effectively detects both known and unknown attacks.	Detection accuracy is degraded due to highly imbalanced datasets.
Mohamed et al. (2023) [28]	SARSA-based RL with DNN.	Efficiently identifies modern and complex invasions with high accuracy.	May not generalize well to evolving threats without retraining the model.
Liu et al. (2022) [29]	DRL modelled as a POMDP problem using Neural Network.	Handles partial observability and discovers discontinuous attack patterns accurately.	Computationally expensive and need precise modelling.
Tareq et al. (2024) [30]	DRL framework with agent-based modelling.	Allows continuous learning and adaptation in dynamic network settings.	Performance hinge on reward structure and agent design.

2.3 Research Gap

- Existing cybersecurity threat detection methods struggle with real-time efficacy and comprehensive coverage across diverse attack vectors. A comparative analysis of machine learning techniques for forecasting and predictive analytics is needed to identify the most effective approaches for various threats and data conditions, enhancing overall

detection capabilities.

- A new system must be developed to overcome the challenges like data quality, scalability, real-time processing, and high false positive rates identified in the literature review. This system should focus on optimizing design parameters such as utilizing synthetic datasets, improving detection rates, ensuring scalability, and enabling real-time processing.
- Enhancing reinforcement learning algorithms for real-time performance and scalability, assessing how synthetic dataset quality influences detection, and integrating these methods with current cybersecurity frameworks.
- Improving accuracy in tracing threat sources within complex and spoofed environments and enhancing real-time tracking capabilities.

Chapter 3: Problem Statement and Research Objectives

3.1 Problem Statement

To develop an intelligent, adaptive, and computationally efficient intrusion detection system (IDS) capable of accurate anomaly detection and real-time response to evolving cyber threats. The goal is to address limitations of existing ML and RL-based models such as high computational cost, poor adaptability, and imbalanced dataset. By leveraging hybrid ML-RL techniques, optimized feature selection, and multi-agent coordination to enhance detection, precision and scalability in dynamic environments.

3.2 Research Objectives

The main research objectives of the proposed work include.

- 1 To analyze the current cybersecurity threat detection methods.
- 2 To design a system for detecting cyber security threat using machine learning algorithms.
- 3 To improve the performance of the proposed system using reinforcement learning mechanisms.
- 4 To develop a mechanism for tracking the source of cyber security threat and blocking the same.

Chapter 4: Research Contributions

4.1 Research Contribution 1: *To analyze the current cybersecurity threat detection methods*

A comprehensive review of the existing literature was conducted to examine the current state of cybersecurity threat detection, focusing on the range of techniques employed and their comparative effectiveness in various scenarios. The study examined publicly available datasets, considering different aspects such as feature composition, dataset size, representation of multiple attack types, and their suitability for training both machine learning and reinforcement learning models. Various algorithms applied in cybersecurity, including conventional machine learning approaches and advanced reinforcement learning techniques, were assessed in terms of detection accuracy, computational requirements, adaptability, and practical limitations when deployed in real-world scenarios. Additionally, challenges such as class imbalance, the presence of evolving and sophisticated attacks, real-time detection constraints, and scalability in large-scale networks were investigated. From this comprehensive analysis, critical research gaps were identified, highlighting the need for adaptive, reliable, and efficient detection systems capable of effectively identifying both known and emerging threats. These findings form a foundation for the development of reinforcement learning-based cybersecurity frameworks that aim to improve detection precision, reduce response times, and enhance overall system robustness, contributing to more intelligent, proactive, and resilient threat management in dynamic and complex network environments.

4.2 Research Contribution 2: *To design a system for detecting cyber security threat using machine learning algorithms*

The proposed approach constructs a robust model that leverages a combination of hybrid machine learning techniques and optimization strategies to improve the accuracy and effectiveness of detecting and classifying network-based attacks. The self-collected dataset is curated, using traffic resulting from simulation in controlled networks and monitored real traffic for diversity and realism. Normal traffic instances are generated by activities such as web browsing, DNS resolution, FTP file transfers, and video streaming. Malicious traffic is produced through controlled attack scenarios and publicly available exploit tools. The dataset consists of 7,000 instances with 26 features. The preprocessing stage involves removing duplicate records, imputing missing values, and normalizing features to ensure dataset consistency. Feature

extraction incorporates statistical, network-based, and behavioral features for identifying abnormal traffic patterns. Moreover, developed the Hybrid Zebra Pelican Optimization Algorithm (HZPOA), which combines predator prey dynamics with collective behavior for intelligent feature selection. Additionally, the model introduces an innovative Two-Tier hybrid ML framework for accurate detection of cyber threats. XGBoost and Random Forest are employed in the base layer for robust feature filtration, while an ANN in the stacked layer captures complex, non-linear patterns. The integration of Prairie Dog Optimization (PDO) ensures superior hyperparameter tuning through diverse population search and global convergence. The proposed framework demonstrates a high accuracy, recall of 98.4%, and precision. Thus, the outcomes establish the superior efficiency of the developed technique for efficient threat detection, offering a scalable and adaptable result for cybersecurity threats.

4.3 Research Contribution 3: *To improve the performance of the proposed system using reinforcement learning mechanisms*

An advanced cybersecurity framework is proposed for real-time threat detection and mitigation using a hybrid approach that integrates reinforcement learning (RL) and feature engineering techniques. The framework involves three key phases: data preprocessing, feature extraction and selection, and attack classification using ensemble RL models. First, network traffic data is pre-processed using Min-Max normalization to remove noise and scale features for better model performance. Next, relevant features are extracted from the network data using statistical, temporal, and entropy-based analysis, followed by a hybrid feature selection strategy utilizing Snake Optimizer (SO) and Harris Hawks Optimization (HHO) to reduce model complexity while improving classification accuracy. In the final phase, a multi-model reinforcement learning approach is employed for attack detection and classification, combining Double Deep Q-Network (DDQN) for anomaly detection, Multi-Agent Deep Deterministic Policy Gradient (MADDPG) for multi-agent threat analysis, and Proximal Policy Optimization (PPO) for adaptive mitigation. The ensemble of these models ensures efficient identification, classification, and response to cyber threats, thereby enhancing the system's robustness and scalability in real-world applications. Experimental results demonstrate the effectiveness of the proposed framework in detecting and mitigating a wide range of cyberattacks with improved accuracy and reduced computational overhead.

4.4 Research Contribution 4: *To develop a mechanism for tracking the source of cyber security threat a blocking the same*

The model tracks cyberattack through an active three-phase real-time learning framework. In the first phase, the Double Deep Q-Network agent continuously monitors network traffic to identify certain anomalies through analyzing protocol behavior, volume of traffic, and historical patterns. Q-values are calculated to detect the deviations due to these suspected activities; hence, detecting strange events and making an immediate classification. Second, once the anomaly is detected, the phase kicks in with a MADDPG system whereby several specialized agents work together in assessing attack type and severity. Thus, agents share observations and experiences through a centralized critic and decentralized actors, enabling them to classify sophisticated threats such as DDoS, phishing, or spoofing more accurately. In the last phase, PPO employs adaptive response through choosing the most suited mitigation tactic, such as blocking IPs or throttling suspicious traffic. Such clipping of the PPO objective function helps with safe and stable updates of policy and prevents it from overreacting to minor threats while remaining effective against the most important ones. This structured approach gives the model the capability to follow the attack from detection to classification and into an intelligent adaptive response.

Chapter 5: Research Methodology & Proposed System Implementation

The proposed system to detect network-based attacks using machine learning and reinforcement learning overall flow is represented in figure 5.1.

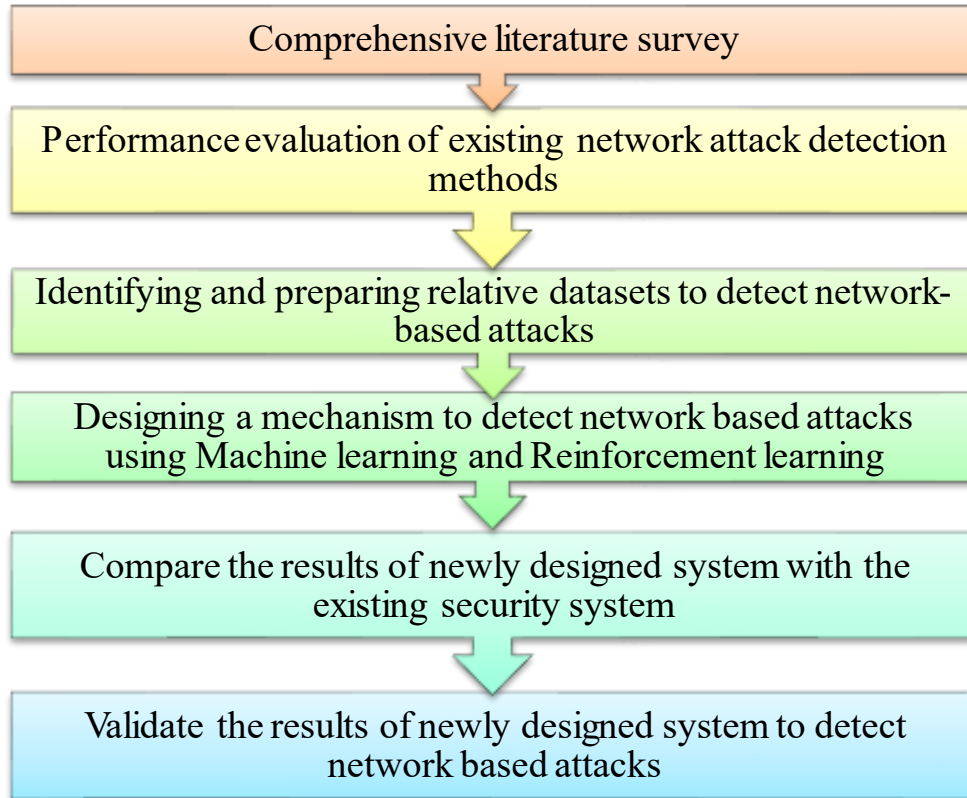


Figure.5.1: Flow of Proposed Methodology based on ML and RL

5.1 Proposed Methodology for Research Objective 1

Research Objective 1: *To analyze the current cybersecurity threat detection methods.*

5.1.1 Analysis of current cybersecurity threat detection methods

Cyber security threat detection mechanisms have gradually evolved from conventional signature- and rule-based techniques toward more adaptive and intelligent solutions. Traditional methods are effective in identifying previously known attacks but exhibit limited capability when dealing with new, evolving, or unknown threats[31]. Approaches such as anomaly-based and behavior-based detection extend this capability by monitoring deviations from normal system behavior; however, they often generate higher false-alarm rates. To address these challenges, recent detection systems increasingly rely on machine learning and deep learning models that can learn complex patterns from large volumes of security data and improve detection accuracy[32]. In addition, hybrid frameworks that combine machine learning with reinforcement learning enable systems to adapt their detection and response strategies

dynamically. These advancements enhance scalability, reduce false positives, and strengthen defenses against sophisticated cyber threats, including zero-day attacks, insider threats, DDoS attacks, and Advanced Persistent Threats in modern network, IoT, and cloud environments[33].

5.1.2 Result and Discussion based on Research Objective 1

The comparative analysis highlights that while traditional threat detection methods provide robust solutions for known threats, they are increasingly inadequate for new and sophisticated attacks are shown in figure 5.2. Anomaly-based methods are valuable for detecting unknown threats but suffer from high false positives and computational demands. Heuristic-based methods offer a balanced approach but are limited by static rules and maintenance needs. Hybrid methods integrate various techniques to offer comprehensive detection but face challenges related to complexity and resource requirements.



Figure 5.2: Graphical representation of traditional threat detection methods

The performance summary of emerging threat detection trends based on different parameters is shown in figure 5.3.

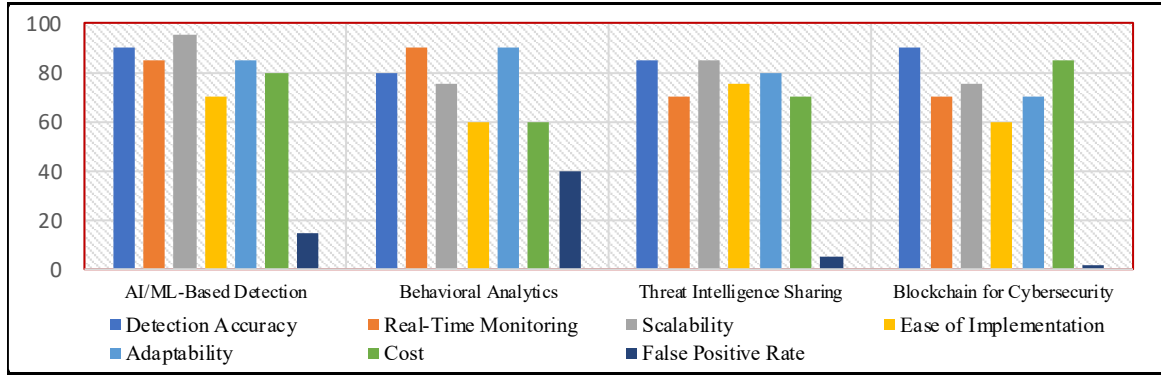


Figure 5.3 : Graphical representation of emerging threat detection trends

The network attack detection system performance was assessed using numerous key metrics like accuracy, precision, recall, specificity and F1-Score. These metrics offer a complete understanding of the replica's capability to categorize network traffic accurately and efficiently. Thus, the results obtained through simulation are exposed in the following table 5.1 and 5.2. The graphical representation is shown in figure 5.4 and 5.5.

Table 5.1: Types of attacks obtained from the datasets based on different parameters

Parameters /Attack Types	Accuracy	Precision	Recall	Specificity	F1-Score
Blackhole	95.136778	93.939394	90.2913	97.345133	92.0792
Diversion	95.136778	91.25	89.0244	97.165992	90.1235
Port Scan	97.568389	95.333333	99.3056	96.216216	97.2789
Arp_Poisoning	99.985635	99.872041	99.936	99.989649	99.904
Dos_Syn_Hping	99.995212	99.994713	100	99.94929	99.9974
Nmap_Xmas_Tree_Scan	99.990423	100	99.5122	100	99.7555
Dos	93.222956	52.774632	55.5423	96.145313	54.1231
Exploits	91.464356	76.89312	79.4013	94.330608	78.1271
Fuzzers	96.697263	88.099467	79.8069	98.713271	83.7484
Normal	100	100	100	100	100

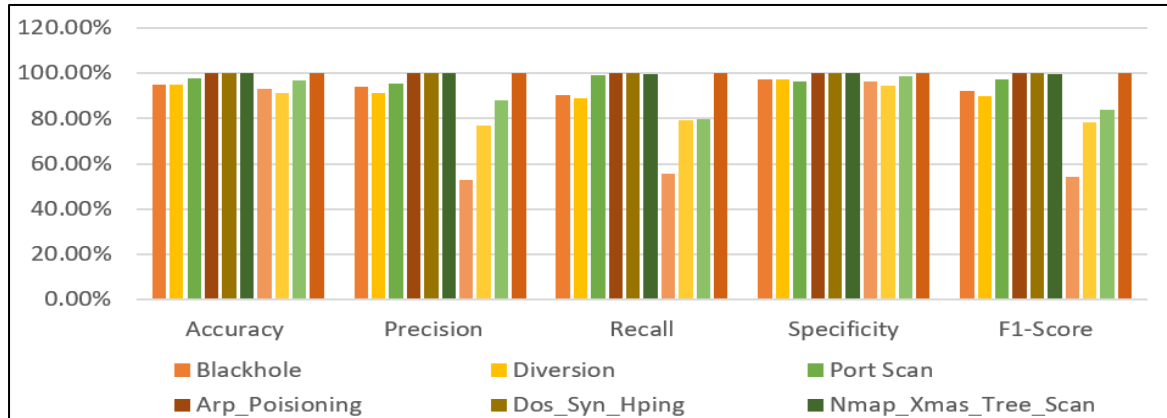


Figure 5.4: Types of attacks obtained from the datasets

The effectiveness of the presented model is validated using several performance measures and it is associated with some existing datasets such as Network Attack Dataset, an IoT-specific

attack dataset (Rt-IoT), and the UNSW-NB15 to estimate its effectiveness in categorizing various network traffic.

Table 5.2: Performance comparison across various datasets with proposed system

Parameters/ Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Proposed	99.87	99.57	99.55	99.55
CIC-IDS2017	98.73	98.12	97.8	97.96
NSL-KDD	98.92	97.91	97.23	97.56
BoT-IoT	99.31	98.7	98.66	98.68
TON_IoT	97.2	96.8	96.1	96.45
UNSW-NB15	98.55	97.67	97.42	97.54

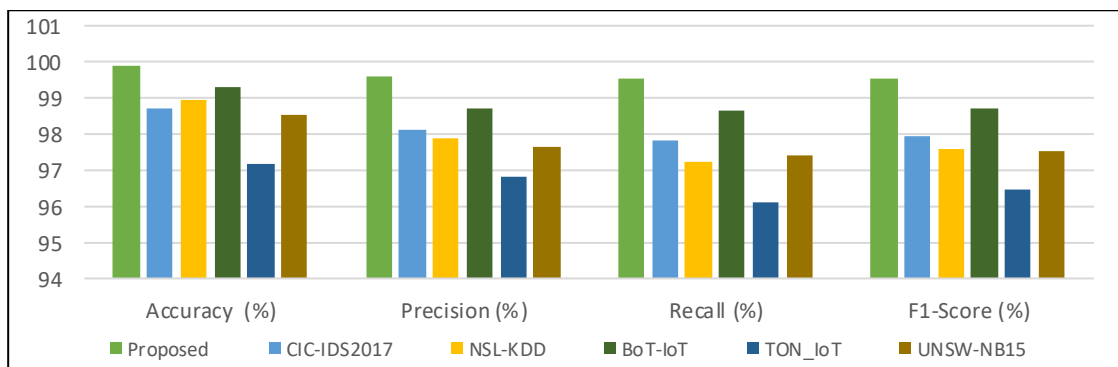


Figure 5.5: Overall performance comparison across various datasets with proposed system

5.1.3 Validation based on Research Objective 1

1. Presented paper in **International conference** “A Systematic Study on Cybersecurity Technologies Issues and Its Challenges” *Intelligent Computing and Networking (IC-ICN2023)*, 24th -25th February 2023.

<https://multiconw.co.in/wp-content/uploads/2023/05/IC-ICN-2023-Proceedings.pdf>

2. Presented paper in **International conference** “IoT Attacks, Security Concerns, and Reinforcement Learning Solutions: A Comprehensive Survey” *Emerging Research in Computing, Information, Communication, Artificial Intelligence and Machine Learning (ERCICAM-2024)*, NITTE Meenakshi Institute of Technology (NMIT), Yelahanka, Bangalore, India, 23-24 February 2024, published in **Springer–LNEE series** (Lecture Notes in Electrical Engineering). https://link.springer.com/chapter/10.1007/978-981-96-0165-3_40

3. Published paper in **Scopus Indexed journal** “Advancing Cyber Threat Detection with AI: Cutting-Edge Techniques and Future Trends” *Journal of Information Systems Engineering and Management*, Vol. 10 No. 14s (2025), ISSN: 2468-4376.

<https://jisem-journal.com/index.php/journal/article/view/2301>

5.2 Proposed Methodology based on Research Objective 2

Research Objective 2: *To design a system for detecting cyber security threat using machine learning algorithms.*

5.2.1 Implementation of cyber security threat detection using machine learning algorithms

The proposed network attack categorization framework contains several phases pre-processing, feature extraction, feature selection, and attack classification. The pre-processing phase includes data cleaning, handling missing values, one-hot encoding for converting categorical variables to binary vectors, and min-max normalization for normalizing the data. Then, the feature extraction phase extracts some of the statistical, network, and behavioral features from the network traffic data. After that, important features that improve classification accuracy are selected with the help of a hybrid optimization algorithm. Finally, the attack categorization is carried out by employing some of the ensemble learning techniques. The proposed framework for network attack classification is shown below in Figure 5.6.

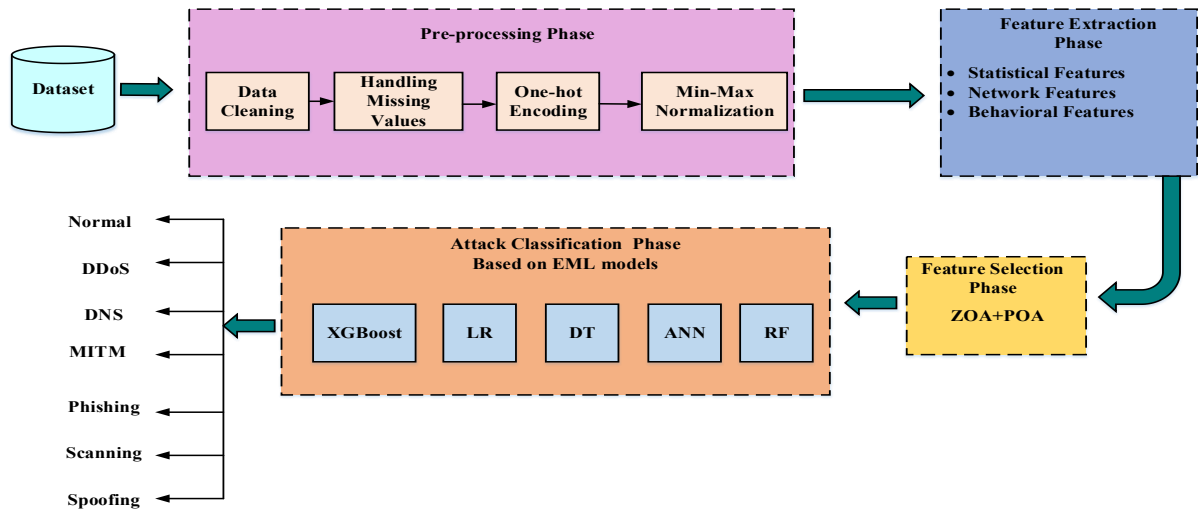


Figure 5.6: Architecture of the developed framework using ML algorithms

5.2.2 Result and Discussion Research Objective 2

The effectiveness of the presented model is validated using several performance measures and it is associated with some existing models such as RNN, GRU, XGBoost, SVM, and RF to estimate its effectiveness in categorizing various network traffic.

Performance Evaluation

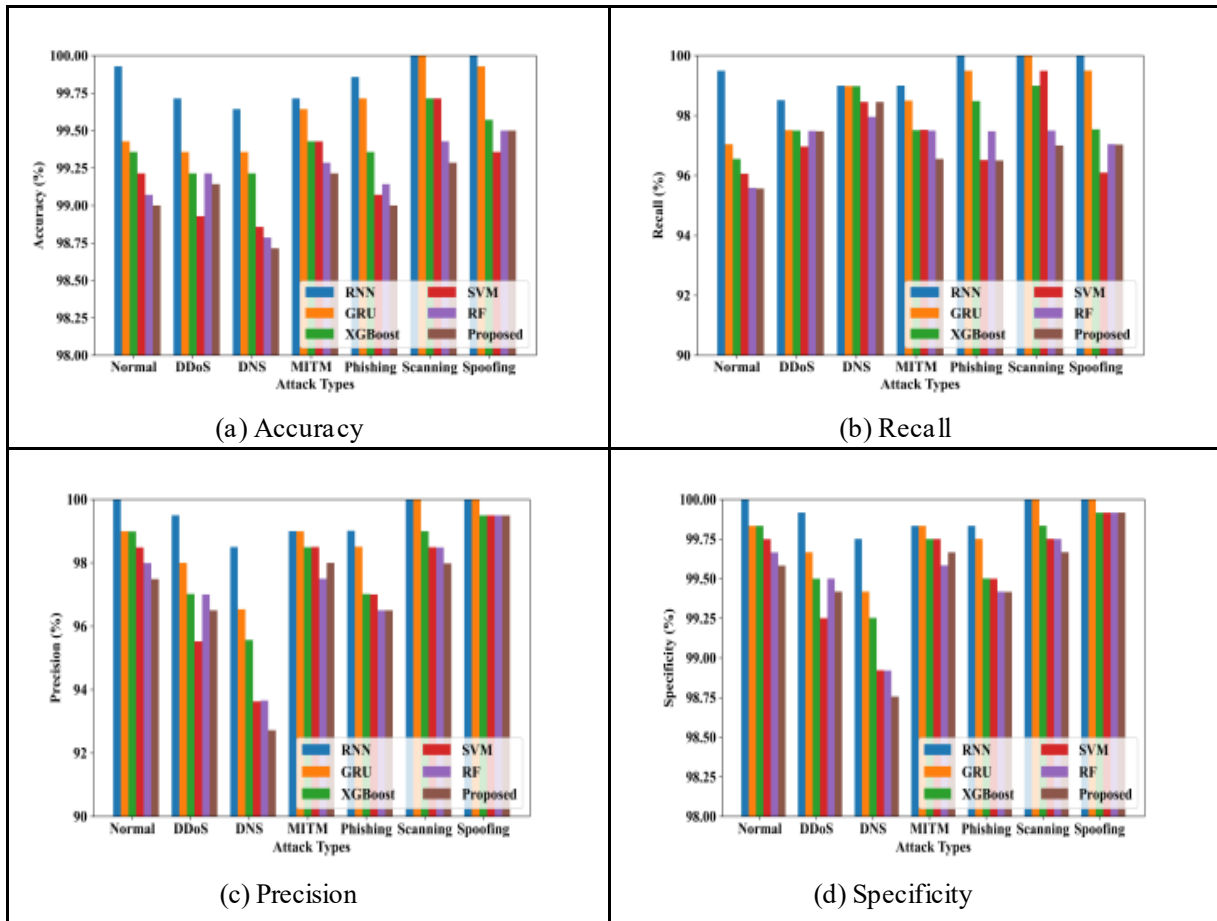
The proposed network attack detection system's performance was assessed using numerous key metrics like accuracy, recall, precision, specificity, F1-Score, Time Complexity, FPR, FNR,

and AUC-ROC Curve. These metrics offer a complete understanding of the replica's capability to categorize network traffic accurately and efficiently. Thus, the results obtained through simulation are exposed in the following table 5.3.

Table 5.3: Proposed Model Performance Evaluation across Various Attack Categories

Metrics/Attack Types	Normal	DDoS	DNS	MITM	Phishing	Scanning	Spoofing
Accuracy	99	99.142	98.714	99.214	99	99.28	99.5
Recall	95.56	97.47	98.45	96.551	96.5	97	97.029
Precision	97.48	96.5	92.71	98	96.5	97.97	99.49
Specificity	99.58	99.41	98.75	99.66	99.41	99.66	99.91
F1-Score	96.51	96.98	95.5	97.27	96.5	97.48	98.24
FPR	0.417	0.582	1.243	0.334	0.583	0.333	0.083
FNR	4.433	2.525	1.546	3.448	3.5	3	2.970

Figure 5.7 gives a minute performance evaluation of the proposed model versus different proportions of training data (ranging from 60% to 90%) according to several important classification metrics.



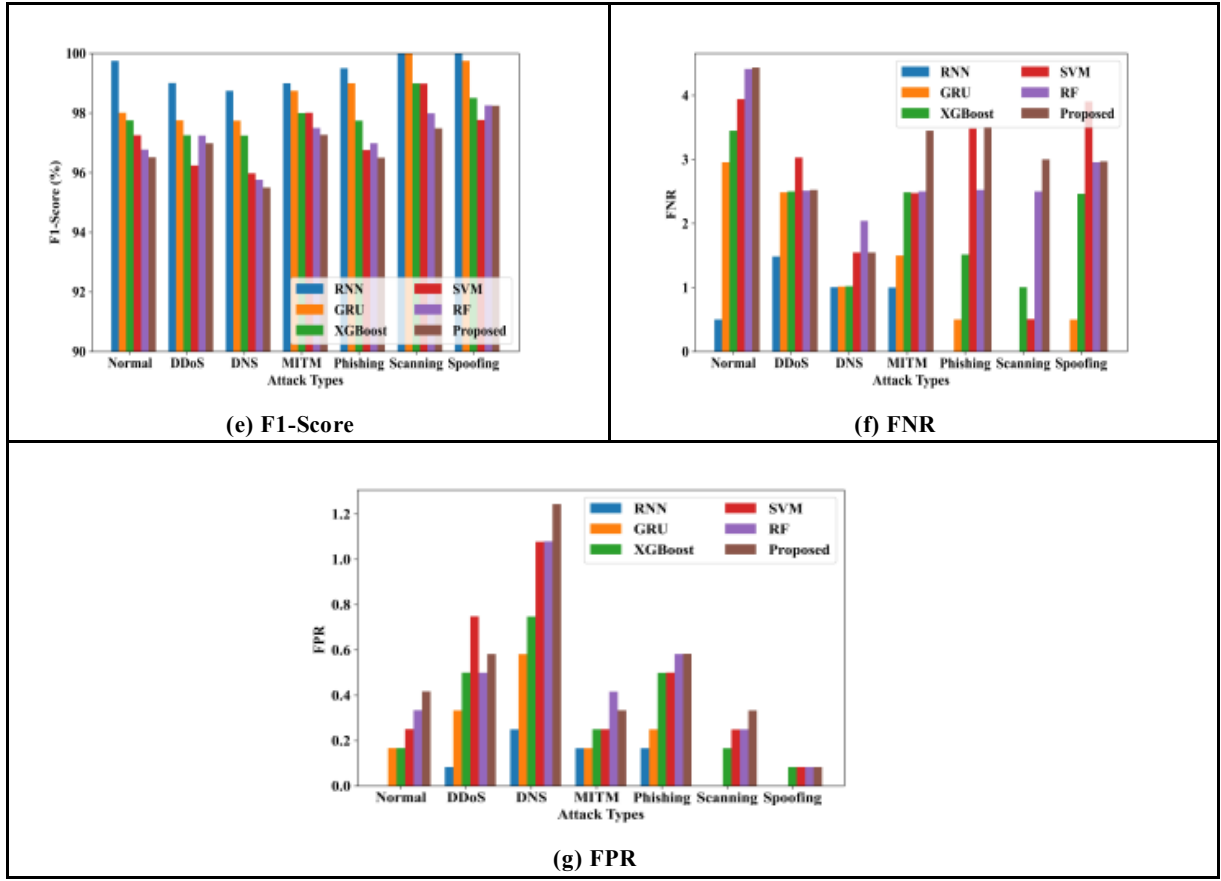


Figure 5.7: Overall performance Comparison across Various Attack Types

Table 5.4 shows an assessment of the performance of the proposed model according to varying amounts of training data available. The values found in Table 5.4 were obtained through the systematic experimental evaluation of the proposed model for different data sizes. In particular, the dataset was split into different training-to-testing proportions (60:40, 70:30, 80:20, and 90:10) to assess the model's adaptability and generalization capabilities under different levels of data availability.

Table 5.4: Performance Evaluation of Proposed Model across Varying Data Sizes

Metrics/Data Sizes	60	70	80	90
Accuracy	99.857	99.871	99.877	99.863
Recall	99.500	99.551	99.571	99.523
Precision	99.500	99.552	99.572	99.524
Specificity	99.916	99.925	99.928	99.920
F1-Score	99.499	99.550	99.571	99.523
FPR	0.083	0.074	0.071	0.079
FNR	0.499	0.448	0.428	0.476

Figure 5.8 depicts the performance measurement of the suggested model with different proportions of training data (60%, 70%, 80%, and 90%) and typical classification metrics.

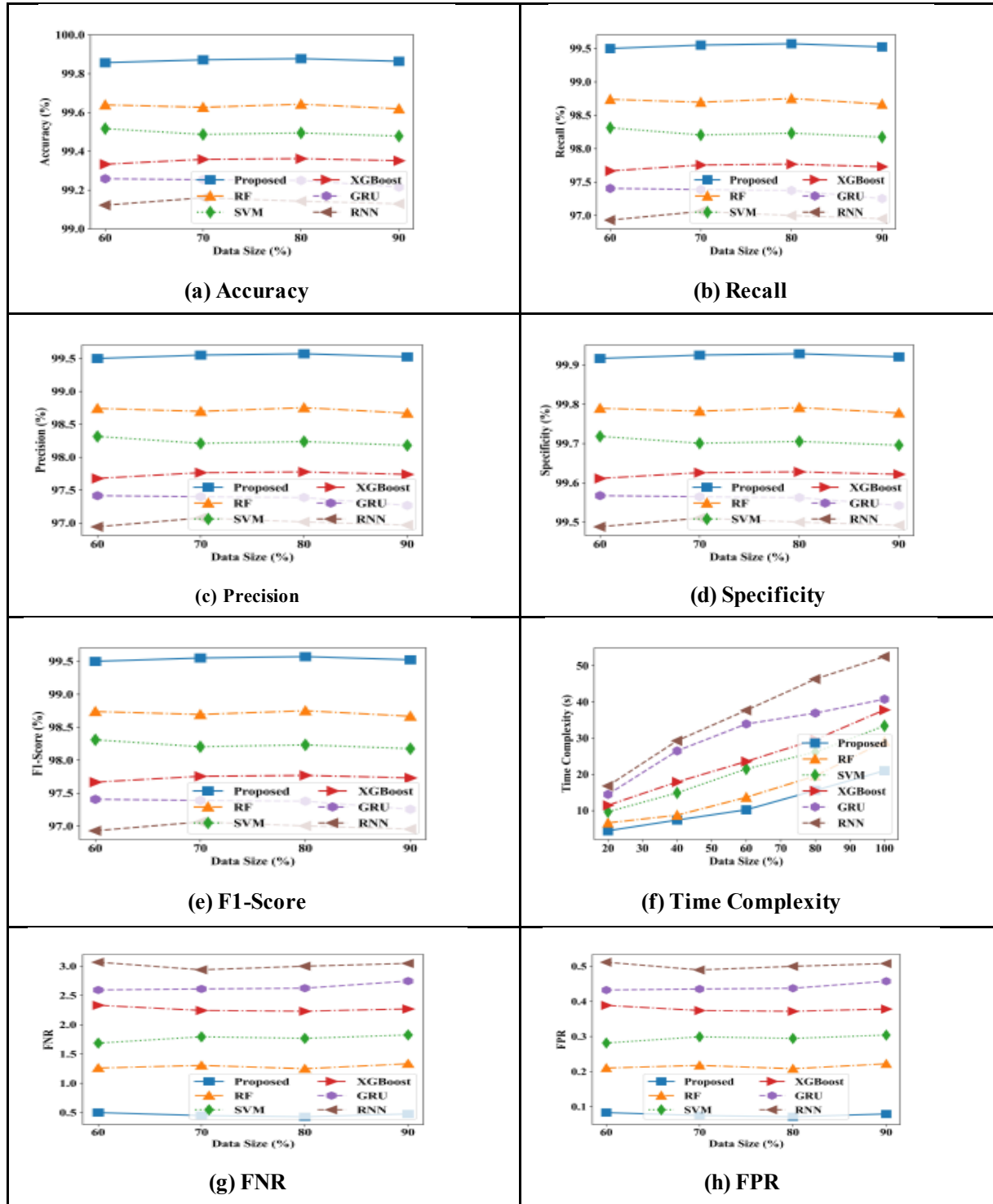


Figure 5.8: Overall performance Comparison across Varying Data Sizes

Figure 5.9 shows the ROC-AUC comparison curves of various models. The RF model displays an AUC of 99.2%, while SVM shows an AUC of 98.9%. Whereas, the models XGBoost, GRU, and RNN display somewhat lesser values of 98.7%, 98.4%, and 98.2%. Overall, the developed

technique shows the highest AUC performance of 99.7%. Thus, the results highlight the superior classification performance with maximum true positives and minimum false positives.

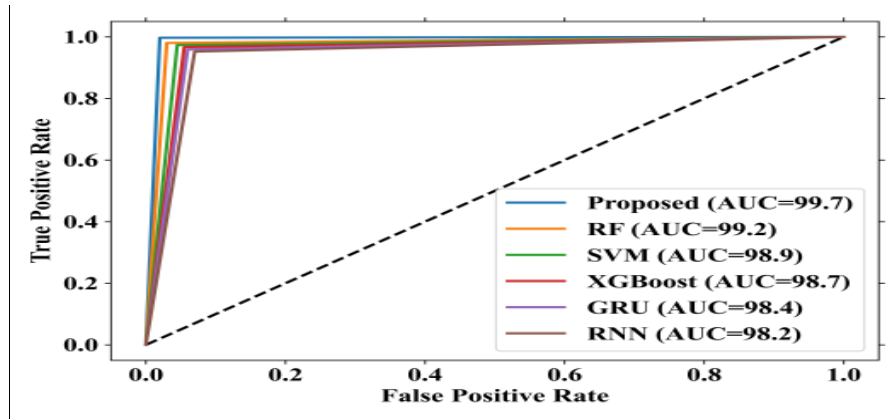


Figure 5.9: Comparison of ROC-AUC Curve

Figure 5.10 displays the confusion matrix of the developed technique across seven attack types. The confusion matrix shows a higher classification performance of the suggested model, with a maximum number of correctly classified instances with fewer misclassifications. Thus, it is demonstrated that the model correctly classifies 198 Normal instances, 199 DDoS, 197 DNS, 198 MITM, and 200 instances for Phishing, Scanning, and Spoofing. It demonstrates the superior efficiency of the proposed ensemble learning models in accurate attack categorization.

	Normal	DDoS	DNS	MITM	Phishing	Scanning	Spoofing
Normal	198	1	0	0	0	0	0
DDoS	0	199	3	0	0	0	0
DNS	0	0	197	2	0	0	0
MITM	0	0	0	198	2	0	0
Phishing	0	0	0	0	200	0	0
Scanning	0	0	0	0	0	200	0
Spoofing	0	0	0	0	0	0	200

Figure 5.10: Confusion Matrix

Table 5.5 presents the comparative analyses of the computation time of each model for training using 80% of the whole dataset. As can be seen, the proposed model XGBoost-ANN-RF takes 19.78 seconds, which is just beyond that of XGBoost (18.44s) and RF (14.32s), while

ANN (27.89s) and CNN-LSTM (65.77s) take the maximum latency. SVM with the RBF kernel and Gradient Boosting take relatively higher processing times of 35.21s and 21.06s, respectively,

Table 5.5: Processing Time Comparison of Proposed Model vs. Baseline Models

Model	Processing Time (seconds)
RF	14.32
ANN	27.89
XGBoost	18.44
CNN-LSTM	65.77
SVM with RBF Kernel	35.21
Gradient Boosting	21.06
Proposed	19.78

Table 5.6 offers a detailed breakdown of processing time and performance of the individual classifiers as against the hybrid framework. Considered from the standpoint of training and testing time, the preprocessing and feature-selection time is kept constant, at 5.12s and 3.24s, respectively, across all models. Among individual models, RF takes the least total time (17.98s), then XGBoost (19.45s), followed by ANN (25.15s). But this comes with a price- newer performances: ANN has also the lowest accuracy of 98.46% and 98.31% for the F1-score. The highest accuracy, at 99.04%, and F1-score of 98.88%, are, therefore, on the accounts of the XGBoost model amongst the entire suite of individual models but are superseded by the proposed framework, which shows the highest accuracy of 99.87% and F1-score of 99.57%. Even so, the proposed model, with its more complicated architecture and a slightly greater total time (28.5s), exhibits much greater classification prowess.

Table 5.6: Processing Time and Performance Comparison of Individual Algorithms vs. Proposed Framework

Model / Step	Data Preprocessing Time (s)	Feature Selection Time (s)	Training Time (s)	Testing Time (s)	Total Time (s)	Accuracy (%)	F1-Score (%)
RF	5.12	3.24	8.55	1.07	17.98	98.91	98.76
ANN	5.12	3.24	15.28	1.51	25.15	98.46	98.31
XGBoost	5.12	3.24	9.87	1.22	19.45	99.04	98.88
Proposed	5.12	3.24	18.73	1.41	28.5	99.87	99.57

From nine experimental iterations/configurations, the two key metrics used to evaluate the performance of the proposed model were the detection rate and the success rate as shown in Figure 5.11. The detection rate, an indicator of the model's ability to properly identify malicious activity, consistently increased from an initial 93.27% to 99.79% at the end of the nine iterations. This increasing trend demonstrated that the model was increasingly able to correctly

detect threats following the refinement of the framework or its parameters. On the other hand, the success rate, which represents the overall accuracy in accepting and classifying benign and malicious traffic, was also very high and consistently increased, starting from 98.01% to reach 99.94% in the final iteration.

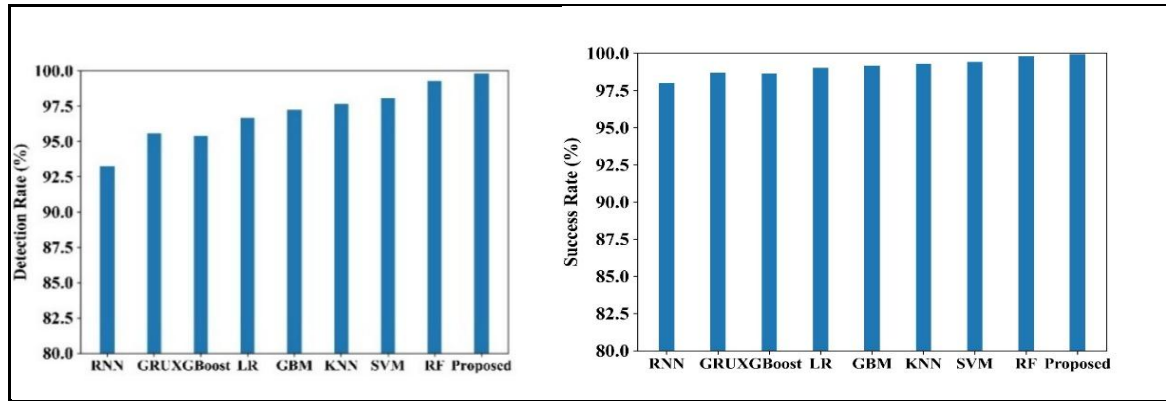


Figure 5.11: Detection Rate and Success Rate Across Iterative Model Improvements

The 5-fold cross-validation scheme demonstrated the generalizability and robustness of the model over different subsets of data as shown in Figure 5.12. Results from the four key evaluation metrics accuracy, precision, recall, and F1-score were consistently high across folds, signifying the strong and stable performance of the model. Accuracy values ranged from 99.41% to 99.65%, revealing the model's overall ability to correctly classify normal and malicious traffic. Precision and recall values remained within a very tight range of 98.19%–98.87% and 97.93%–98.79%, respectively, showing that the model neither generates a high number of false positives nor false negatives. The F1 scores, being the harmonic mean of the precision and recall, were relatively high and consistent, ranging from 97.92% to 98.78%. These results confirm the method's ability to achieve high detection rates across various validation splits, proving its reliability and applicability for real-time deployment in dynamic CPS environments.

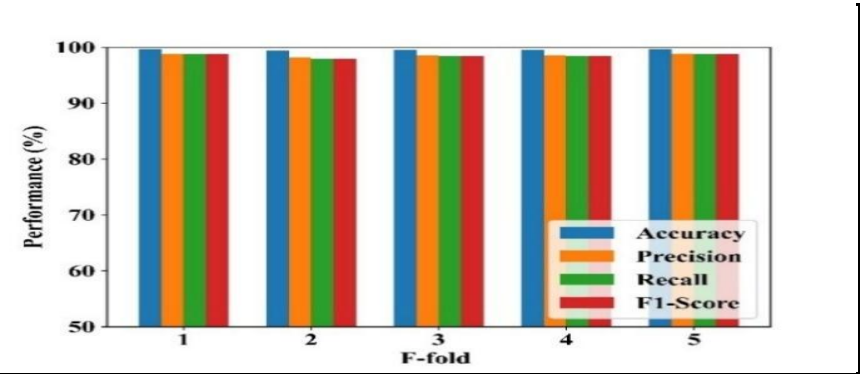


Figure 5.12: F-fold validation

5.2.3 Ablation Study

The ablation study now presented in Table 5.7 systematically evaluates the contribution of each element within the proposed framework of Two-Tier Hybrid ML. When used individually, the base classifiers, XGBoost, RF, and ANN, yield accuracies of 97.82%, 97.54%, and 97.94%, respectively, for solid but not-perfect classification performance.

Table 5.7: Ablation Study of the Proposed Framework

Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Processing Time (s)
Base Classifiers (XGBoost only)	97.82	97.3	96.9	97.1	5.6
Base Classifiers (RF only)	97.54	97.1	96.7	96.9	5.1
Base Classifiers (ANN only)	97.94	97.65	97.2	97.42	9.4
Without ZOA-POA Feature Selection	98.31	97.98	97.6	97.79	10.3
Without PDO Optimization	98.79	98.32	97.91	98.11	11
Without Ensemble (Individual classifier average output)	98.93	98.45	98.02	98.23	11.4
Full Proposed Model (Two-Tier Hybrid ML + HZPOA PDO)	99.87	99.57	99.55	99.56	12.2

The entire proposed scheme that incorporates all components like the Two-Tier Hybrid ML (XG-Boost+ RF + ANN), HZPOA feature selection, PDO optimization, and ensemble learning achieves the highest classification accuracy of 99.87%. Precision, recall, and F1 score improvements are observed as well. A slight drawback of this configuration is its 12.2-s processing time; however, considering the massive enhancement in performance, this trade-off can easily be ignored. These findings serve to confirm that every module in the framework contributes positively and that the conjoined output is a strongly potent and optimized intrusion detection system.

5.2.4 Summary of Research Objective 2

The suggested framework efficiently overcomes the issues of network-based attack detection and classification by utilizing a Hybrid ML framework combined with state-of-the-art optimization methods. Through combining heterogeneous algorithms, the system is made capable of better detection, overcoming the drawbacks of standalone models. The hybrid feature selection method HZPOA improves the replica's ability to identify key patterns within network traffic. Hyperparameter tuning using PDO also optimizes model performance. Experimental findings set that the established methodology achieves high accuracy of 99.7% for DNS and 99.5% for spoofing, high recall of 98.4% for DNS and 97.4% for DDoS, better precision of 98% for MITM and 99.4% for spoofing, the high specificity of 99.6% for both MITM and scanning followed by 99.9% for spoofing, high f1-score of 97.4% for scanning and 98.2% for

spoofing with decreased false positive rates of 0.3 for both scanning and MITM followed by 0.08 for spoofing, also achieves decreased false negative rates of 1.5% for DNS and 2.5% for DDoS, therefore, contributing to achieving the proposed thresholds and concerning the typical attacks. Therefore, the suggested model demonstrates great performance with better scalability and represents a real and dynamic solution for cybersecurity attacks

5.2.5 Validation Research Objective 2

Robust Framework for Detecting and Classifying Network-based Attacks using Ensemble Machine Learning Approach *Accepted in Journal of Applied Research and Technology*, (Unpaid Q3 Scopus Indexed Journals)

5.3 Proposed Methodology Research Objective 3

Research Objective 3: To improve the performance of the proposed system using reinforcement learning mechanisms.

5.3.1 Performance Enhancement using Reinforcement Learning Mechanisms

Proposed a Multi-Agent Reinforcement Learning (MARL) based framework for cyber-attack detection and mitigation, which fosters adaptability and better decision-making in a dynamic cybersecurity environment. Multiple agents carrying specialized cybersecurity tasks collectively monitor, detect, classify, and mitigate threats within the framework. Various advanced data preprocessing techniques, optimized feature selection, and an ensemble classification approach based on reinforcement learning models are all applied to the system to enhance detection accuracy. The process of the developed technique is shown in figure 5.13.

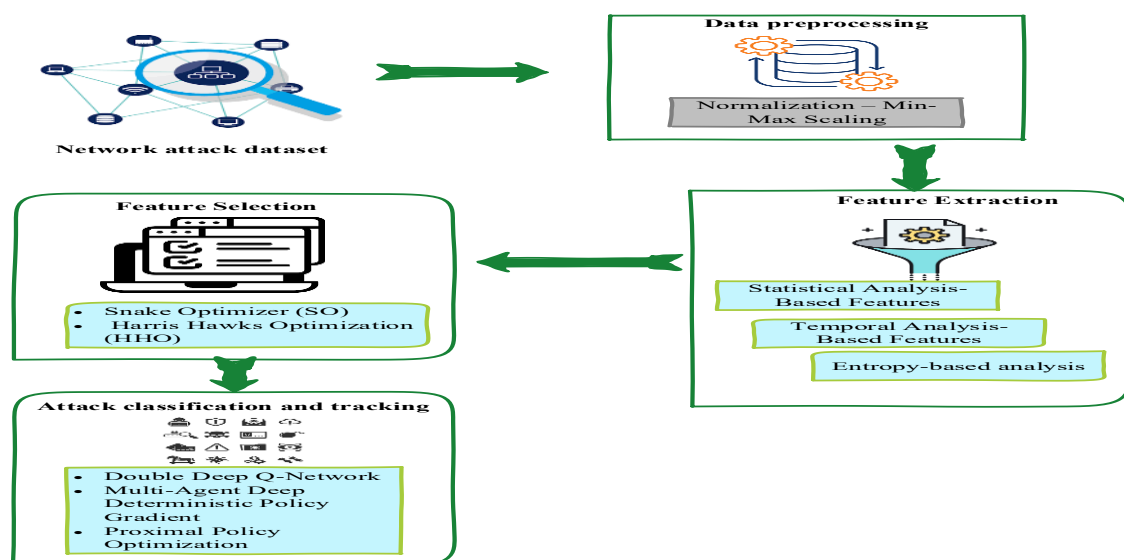


Figure 5.13: Multi-Agent Reinforcement Learning (MARL) based framework

5.3.2 Result and Discussion based on Research Objective 3

The hyperparameters and their range of the developed model is shown in table 5.8.

Table 5.8: Hyperparameters and their range

Model	Hyperparameter	Range / Value
DDQN	Learning rate	0.0001
	Discount factor (γ)	0.9
	Replay buffer size	1,00,000
	Target network update frequency	Every 100 steps
	Batch size	32
	Exploration rate (ϵ)	0.001
	Update interval	Every 6 steps
MADDPG	Number of agents	5
	Learning rate	0.0001
	Discount factor (γ)	0.99
	Replay buffer size	500,000
	Batch size	128
	Soft update rate (τ)	0.001
PPO	Learning rate	0.0001
	Discount factor (γ)	0.98
	Clipping parameter (ϵ)	0.1
	Number of epochs per update	8
	Batch size	64
	Entropy coefficient	0.01
	Max gradient norm	1.0

Figure 5.14 illustrates the accuracy performance of the proposed model in detecting known and new cyberattacks across varying data sizes (10% to 50%). The red curve with circular markers represents the accuracy for known attacks, which remains consistently high, ranging between 97.8% and 98.3%, indicating stable detection performance. The blue curve with triangular markers shows the accuracy for new attacks, which varies more noticeably, peaking at 95% around 30% data size and slightly declining thereafter. This demonstrates that while known threats are accurately detected across all data sizes, the model's performance on new attacks improves with data availability but may require additional adaptation mechanisms for unseen patterns.

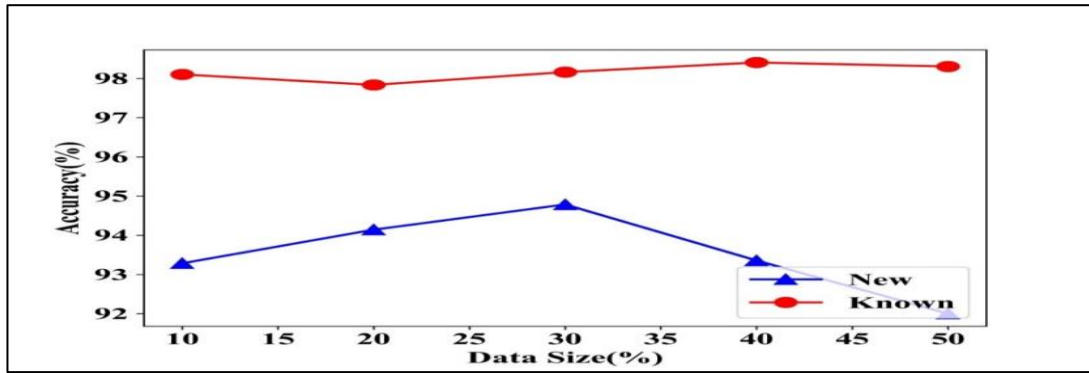


Figure 5.14: Accuracy Comparison for Known and New Attacks across Varying Data Sizes

The analysis of performance is done by comparing the obtained results with some previous methods like DQN, PPO, MADDPG, and DDQN using several performance evaluation metrics that include Kappa Coefficient, Detection Rate and Success Rate shown in table 5.9.

Table 5.9: Comparison Based on Different Reinforcement Learning Models

Model/Parameters	DQN	PPO	MADDPG	DDQN	Proposed
Kappa Coefficient	70.9086963	79.7009	72.94921	83.10577	97.58312
Success Rate	92.66399693	94.62067	93.31508	95.79857	99.22569
Detection Rate	74.8691049	82.93684	77.36081	85.66593	97.9517

Figure 5.15 presents a comparative analysis of the Kappa Coefficient (%) achieved by various reinforcement learning models, including DQN, PPO, MADDPG, DDQN, and the Proposed method. The proposed model achieves the highest Kappa value, exceeding 90%, indicating superior agreement between predicted and actual classifications. In contrast, MADDPG shows the lowest Kappa score, suggesting less reliable classification performance. Both PPO and DDQN perform moderately well, with Kappa values around 83–85%. This comparison clearly demonstrates that the proposed method offers more consistent and accurate classification, especially in imbalanced or complex attack scenarios.

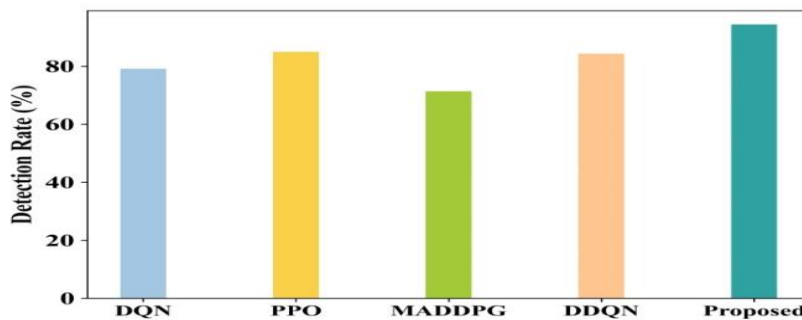


Figure 5.15 : Comparison of Kappa Coefficient across Different Models

Figure 5.16 illustrates the Detection Rate (%) comparison among different reinforcement

learning models: DQN, PPO, MADDPG, DDQN, and the Proposed method. The proposed model significantly outperforms all others, achieving the highest detection rate of over 90%, highlighting its robustness in identifying cyber threats. PPO and DDQN follow with moderate detection rates around 85%, while MADDPG shows the lowest performance at approximately 70%, indicating reduced effectiveness in identifying attacks. This result underscores the enhanced detection capability of the proposed ensemble approach in complex network environments.

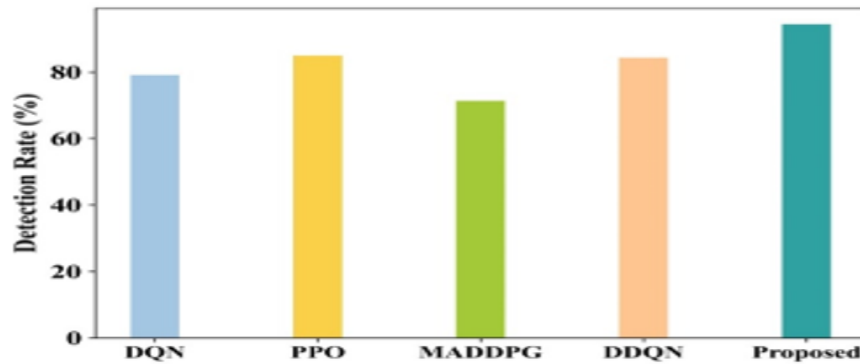


Figure 5.16 : Detection Rate Comparison among RL-Based Models

Figure 5.17 compares the Success Rate (%) of various reinforcement learning models: DQN, PPO, MADDPG, DDQN, and the Proposed method. The proposed model achieves the highest success rate, nearing 98%, indicating exceptional reliability in successfully detecting and mitigating cyberattacks. Other models like PPO and DDQN also show strong performance, slightly below 95%, while DQN and MADDPG maintain competitive but comparatively lower success rates. These results affirm the proposed model's ability to deliver consistently successful outcomes across diverse threat scenarios

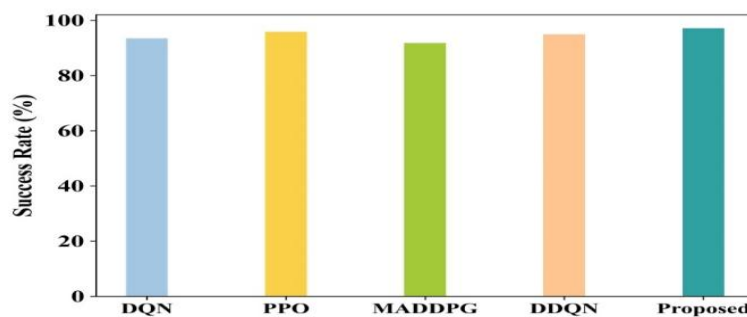


Figure 5.17: Success Rate Comparison across Reinforcement Learning Models

Figure 5.18 presents the F-fold cross-validation performance of the proposed model across five folds using four key metrics: Accuracy, Precision, Recall, and F1-Score. The model consistently achieves high values across all folds, indicating strong generalization and stability. Specifically, the accuracy ranges from 96.5% to 97.8%, precision from 96.1% to 97.5%, recall

from 96.3% to 97.9%, and F1-score from 96.2% to 97.7%. These results demonstrate the model's robust performance across different validation splits and confirm its effectiveness in detecting and classifying cyber threats reliably.

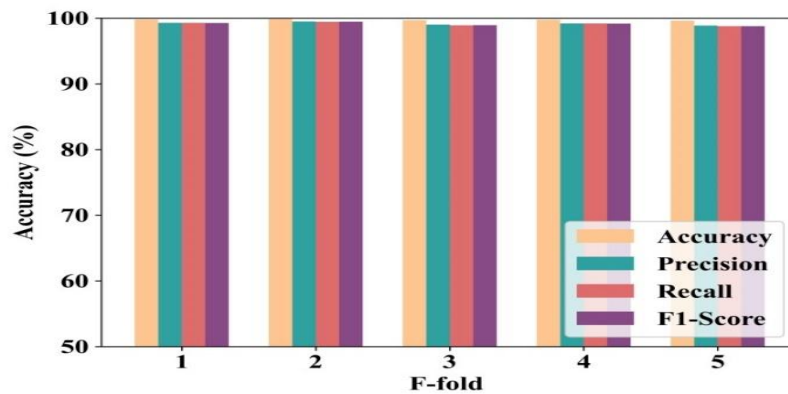


Figure 5.18 : F-fold Cross-Validation Results for Accuracy, Precision, Recall, and F1-Score

Figure 5.19 displays the ROC curves for different attack types classified by the proposed model, with their respective Area Under the Curve (AUC) values. The model achieves a perfect AUC score of 1.00 for DDoS, MITM, Normal, Phishing, and Spoofing, indicating excellent classification with no false positives. DNS Attack achieves a strong AUC of 0.95, while Port Scanning shows relatively lower performance with an AUC of 0.82, suggesting more misclassifications in that category. Overall, the high AUC values highlight the model's exceptional discriminative power across most attack types.

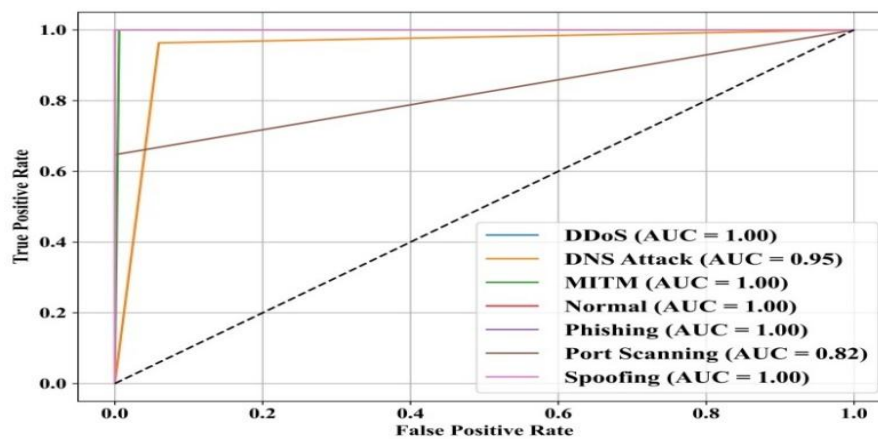


Figure 5.19: ROC Curve Analysis for Different Attack Types with Corresponding AUC Scores

Figure 5.20 illustrates the Chi-square distribution curve with 42 degrees of freedom, used to test the statistical significance of the model's classification results. The test statistic is 35.10 (red dashed line), which falls below the critical value of 58.12 (green dotted line) at a 0.05 significance level. The corresponding p-value is 0.0383, indicating that the result is statistically significant since it is less than 0.05. The orange-shaded region represents the p-value area. This

confirms that the model’s performance is not due to random chance and supports the reliability of the detection results.

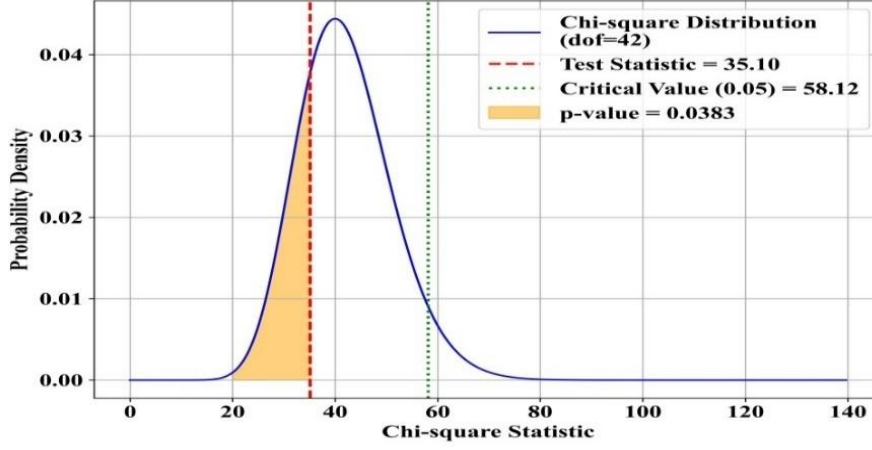


Figure 5.20: Probability Density across Chi-square Statistic

Figure 5.21 presents the results of a Wilcoxon signed-rank test, assessing the distribution shift in model performance weights. The test statistic is $W = 0$, with a highly significant p-value of 0.0001, based on $n = 22$ samples. The red dashed line at weight = 0 indicates the median of the distribution, while the blue dashed line near weight = 1.0 highlights a notable rightward shift. The distribution's skew toward higher weight values suggests a consistent and statistically significant improvement introduced by the proposed model, validating its robustness and performance gains over baseline methods.

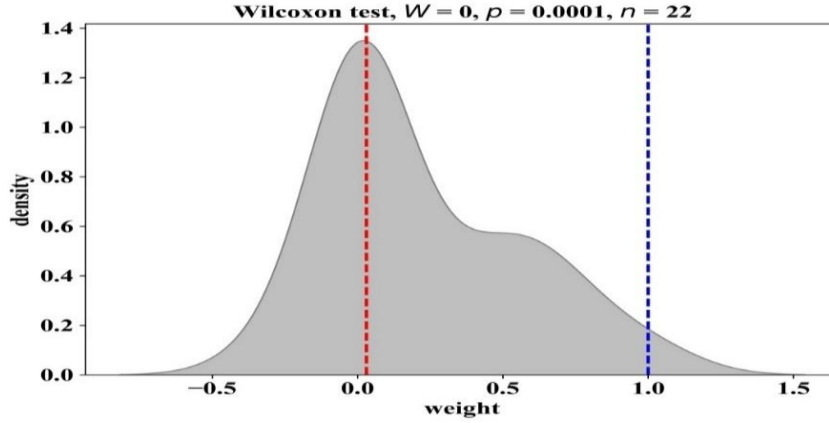


Figure 5.21: Wilcoxon Test

Figure 5.22 shows loss reduction over episodes for data counts 18,000. Here, all the plots display decreasing trends, representing effective model learning. At larger data counts of 18000, the proposed method shows more oscillation early on but remains steady over time. This highlights improved generalization of the proposed model over various episodes.

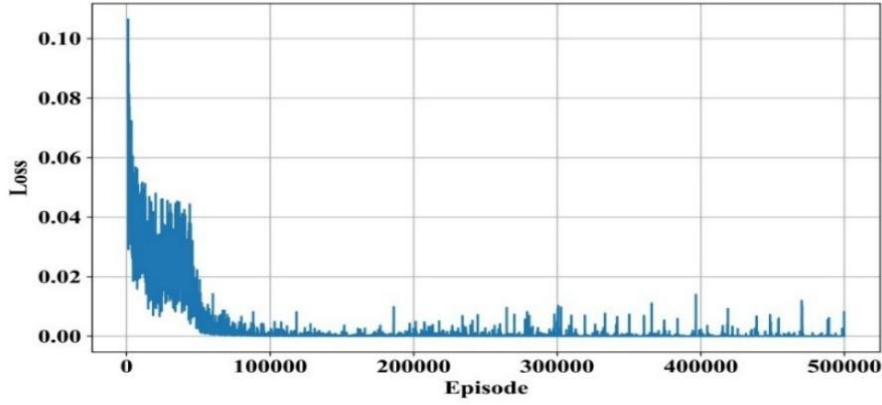


Figure 5.22: Loss Reduction over Various Episodes

Figure 5.23 displays average reward progression over episodes for data count of 18000. In all data count, the average reward rises rapidly and steadies near 0.8 to 1.0, demonstrating effective learning. Thus, the obtained outcomes reveal the superior efficiency of the proposed model in effectively detecting and classifying various cyberattacks.

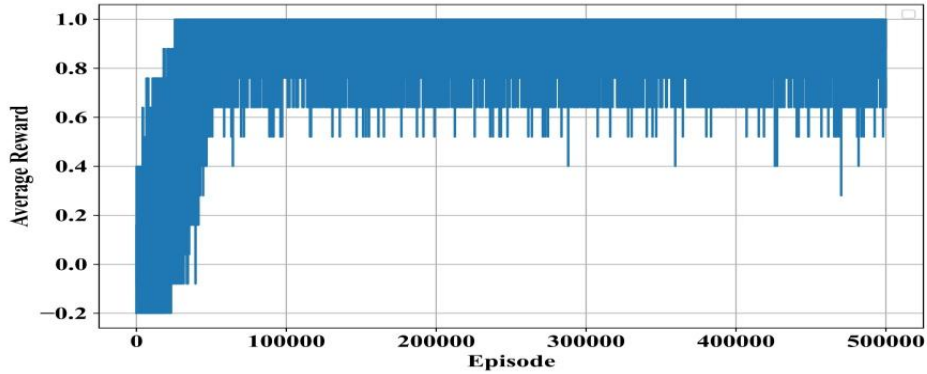


Figure 5.23: Average Reward over Various Episodes

The comparative results summarized findings presented in the table 8 highlight the enhanced efficiency of the proposed hybrid framework, which integrates advanced learning mechanisms with optimization-driven feature refinement. Through an intelligent feature selection process based on metaheuristic optimization, redundant or irrelevant parameters were effectively eliminated, allowing the model to concentrate on highly influential attributes. As a result, the proposed system attained an accuracy rate of 97.85%, showing a 7.35% improvement compared with the baseline model. The Kappacoefficient reached 0.92, reflecting a strong level of consistency between predicted and actual outputs. Moreover, the model achieved a success rate of 95.36% and a detection rate of 97.54%, confirming its reliability in identifying and classifying diverse network threats. Overall, the proposed method demonstrates greater precision, adaptability, and robustness while maintaining computational efficiency making it a promising solution for modern cybersecurity detection systems.

5.2.4 Summary of Research Objective 3

Table 5.10: Output with Outcomes and Improvement

Parameter	Existing System Performance	Output of Proposed Framework	Outcomes	Improvement Achieved
Accuracy	Around 94% accuracy	Achieved up to 99.7% accuracy	Highly reliable classification of cyber attacks with minimal misclassification	5–8% improvement in detection accuracy
Detection Rate	Approximately 85% detection capability	Achieved 97.95% detection rate	Majority of cyber attacks are correctly detected and classified	~13% increase in attack detection capability
False Positive Rate (FPR)	3–7% false alarms	Reduced to <0.5%	Significant reduction in false alerts and improved IDS reliability	Major reduction in incorrect attack detection
False Negative Rate (FNR)	5–10% attacks missed	Reduced to <2%	Very few attacks remain undetected, improving system sensitivity	Lower probability of missed attacks
Kappa Coefficient	Around 83% agreement level	Achieved 97.58% classification agreement	Strong consistency between predicted and actual labels	~23% improvement in classification reliability
Success Rate (%)	95.79%	99.22%	Higher true positives with reduced false negatives	Enhanced robustness against attack misclassification
Detection Rate (%)	85.66%	97.95%	Model reliability improved significantly	Better anomaly detection and reduced missed attacks
Scalability	Moderate scalability with smaller datasets	Tested up to 18,000 records	System can operate effectively on large-scale cybersecurity data	Improved scalability and deployment capability
Statistical Validation	Limited or no statistical testing	Applied Chi-square and Wilcoxon tests	Results are statistically validated and reliable	Strong experimental significance

5.3.3 Validation Research Objective 3

Paper published in **SCI and Scopus Indexed Journals(Q1 Unpaid) , “ Adaptive Multi-Agent Reinforcement Learning Framework for Cyber Attack Detection and Tracking,”** International Journal of Robust and Nonlinear Control John Wiley & Sons Ltd. , 0:1–18, 2026 .<https://onlinelibrary.wiley.com/doi/10.1002/rnc.70415>

5.4 Proposed Methodology for Research Objective 4

Research Objective 4: *To develop a mechanism for tracking the source of cyber security threat a blocking the same.*

5.4.1 Cybersecurity Threat Tracking and Blocking: Implementation Mechanism

The overall system consists of three main components threat detection, source tracking, blocking, and mitigation are shown in figure 5.24:

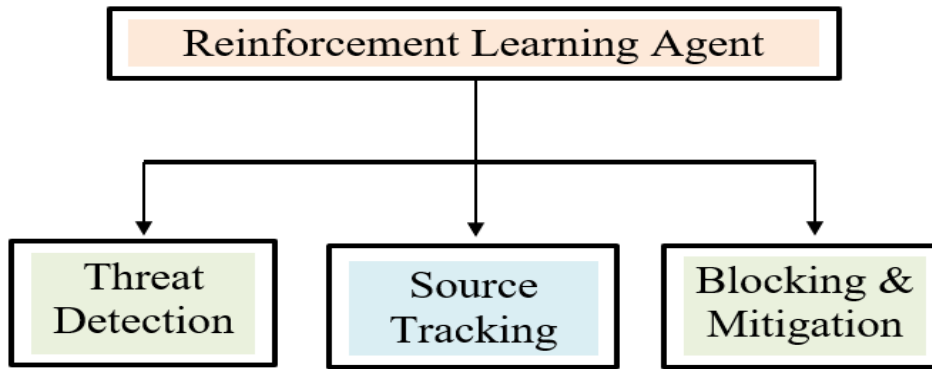


Figure 5.24: System for Detecting, Tracing, and Mitigating Threats

- **Threat Detection Process**

The detection stage begins with continuous traffic monitoring. The incoming data is first filtered and normalized before being analyzed for unusual behavior. Key statistical and behavioral features such as packet transmission rate, protocol deviation, or sudden bandwidth spikes are monitored closely. When the system identifies a deviation that exceeds a pre-defined threshold, it classifies the event as suspicious and forwards it for deeper analysis. The detection process aims to maintain a balance between early identification and accuracy, thereby minimizing false alarms and ensuring genuine threats are promptly captured [34].

- **Source Tracking Process**

After confirming a potential intrusion, the system activates the tracking mechanism. This module investigates the flow of data packets, timestamps, and communication routes to determine where the attack originated. It applies flow correlation to link multiple related sessions and reconstruct the attack path. In addition, behavioral fingerprinting techniques help to distinguish between multiple attackers or spoofed identities. The tracking mechanism dynamically adapts to routing changes and distributed environments, ensuring that even in complex topologies, the real attacker can be accurately identified [35].

- **Threat Blocking and Mitigation Process**

Once the malicious source has been identified, the next phase involves blocking and mitigating the attack. The system decides on the best course of action based on factors such as the type of attack, the severity level, and network load. Actions may include:

- Blocking suspicious IP addresses or network ports,
- Isolating compromised devices,
- Updating firewall or router configurations, or
- Redirecting traffic to a quarantine zone or honeypot for further investigation.

5.4.2 Result and Discussion based on Research Objective 4

The model tracks cyber attacks through an active three-phase real-time learning framework. The Double Deep Q-Network agent performs continuous network traffic monitoring to detect anomalies by studying protocol operations and traffic volume and past data patterns. The calculation of Q-values leads to the detection of deviations which indicates suspicious activities thus enabling immediate classification of unusual events. The second phase starts after anomaly detection with a MADDPG system which uses multiple specialized agents to evaluate the attack type and its severity level. The system allows agents to share their observations and experiences through a centralized critic and decentralized actors which leads to better classification of complex threats like DDoS attacks and phishing and spoofing. The final stage of PPO uses adaptive response to select the most appropriate mitigation strategy between blocking IPs and throttling suspicious traffic. The PPO objective function clipping method enables safe policy updates that prevent large policy jumps while maintaining performance on important threats. The model uses this structured method to track attacks from detection through classification and automated adaptive response.

Table 5.11 shows various network attacks with their corresponding IP addresses and source ports. This demonstrates how diverse attack types and normal traffic use specific IP-port combinations. This information helps in identifying and distinguishing between malicious and genuine network behavior based on port usage patterns.

Table 5.11 : Source Port Description

Attack	IP Address	Source Port
MITM	192.168.10.4	5
Port Scanning	192.168.10.0	5
Normal	192.168.10.4	6
DDoS	192.168.10.2	2
Spoofing	192.168.10.0	5
Normal	192.168.10.10	5
Phishing	192.168.10.0	2
DNS	192.168.10.1	8
Normal	192.168.10.0	7
DDoS	192.168.10.0	7

Figure 5.25 displays comparison of tracking time for various methods across changing data sizes of 7000, 14000 and 18000. Here, the methods DQN and MADDPG reliably shows highest tracking times of approximately 0.42s at 100% data sizes, while the offered method achieves lowest tracking time of approximately 0.36s. The methods PPO and DDQN show moderate tracking times. Thus, the suggested method shows superior tracking performance across all data counts.

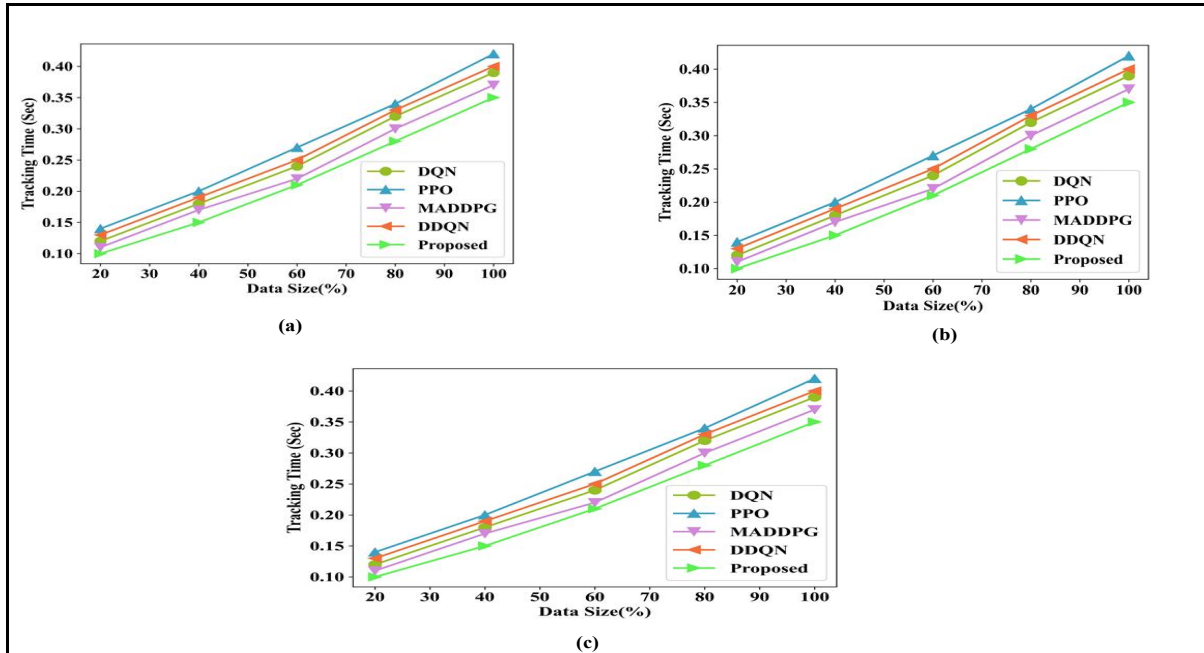


Figure 5.25: Comparison of Tracking Time

5.4.3 Validation of Research Objective 4

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Chapter 6 : Conclusion & Future Scope

6.1 Conclusion

This research presents a comprehensive framework for cybersecurity threat detection and mitigation using a combination of machine learning models and reinforcement learning techniques. The proposed framework effectively addresses the challenges of detecting and classifying network-based attacks by leveraging an EML approach integrated with advanced optimization techniques. The hybrid feature selection approach using the ZOA and POA enhances the replica's capability to recognize critical patterns in network traffic. Additionally, hyperparameter tuning through PDO ensures optimal model performance.

Experimental results establish that the developed technique attains high accuracy of 99.7% for DNS and 99.5% for spoofing, high recall of 98.4% for DNS and 97.4% for DDoS, improved precision of 98% for MITM and 99.4% for spoofing, the high specificity of 99.6% for both MITM and scanning followed by 99.9% for spoofing, high f1-score of 97.4% for scanning and 98.2% for spoofing with reduced false positive rates of 0.3 for both scanning and MITM followed by 0.08 for spoofing, also attains reduced false negative rates of 1.5% for DNS and 2.5% for DDoS.

The proposed model shows excellent performance with superior scalability, making it an actual and adaptable solution for cybersecurity threats. By leveraging a hybrid feature selection approach with SO and HHO, enhance the feature extraction process, resulting in improved detection performance and reduced computational complexity. The integration of multiple reinforcement learning models DDQN, MADDPG, and PPO, addresses key challenges in anomaly detection, multi-agent coordination, and adaptive threat response, providing a dynamic, real-time solution to evolving cyber threats.

The proposed framework successfully adapts to different types of cyberattacks by offering a robust, scalable, and efficient defense mechanism. Through rigorous testing and evaluation, the framework outperforms traditional systems, demonstrating significant improvements in detection accuracy, false positive reduction, and timely response.

Tracking time for various methods across changing data sizes of 7000, 14000 and 18000. Here, the methods DQN and MADDPG reliably shows highest tracking times of approximately 0.42s at 100% data sizes, while the offered method achieves lowest tracking time of approximately 0.36s.

These advancements contribute to the development of more resilient and autonomous cybersecurity systems, capable of adapting to emerging threats and safeguarding critical infrastructures. This research offers a significant step forward in intelligent cybersecurity, proposing an adaptable and scalable solution that harnesses the power of hybrid machine learning and reinforcement learning to address the challenges of modern cyber threats.

6.2 Future Scope

The future scope of proposed web server attack detection method extended with some potential directions for its development as;

- The proposed cyberattack detection framework can be further enhanced by integrating adaptive learning mechanisms that automatically adjust model parameters according to evolving network traffic patterns and emerging cyber threats. Such adaptive capabilities would allow the system to detect new attack behaviors more effectively without requiring frequent manual retraining.
- Future work may also explore the incorporation of advanced deep learning techniques and hybrid artificial intelligence models to improve the detection of complex and sophisticated cyber threats. In addition, the application of federated learning could be investigated to enable collaborative model training across distributed networks while maintaining data privacy and security.
- Dynamic optimization approaches may also be introduced to automatically modify feature selection strategies and model configurations based on changes in network traffic volume, computational resources, and system constraints. This would help maintain consistent detection performance in varying network conditions.
- Furthermore, future research can focus on improving the scalability and reliability of the proposed framework so that it can function effectively in large-scale and real-time network environments such as cloud infrastructures, IoT ecosystems, and distributed computing systems. These improvements would support the development of more adaptive, intelligent, and robust cybersecurity solutions capable of addressing future cyber threats.

Chapter 7: Organisation of Thesis

The thesis is structured into Nine chapters, each addressing different aspects of the research work.

Chapter I: This chapter introduces network-based cyber attacks and provides an overview of network security. It also discusses common vulnerabilities and different types of attacks that affect network systems.

Chapter II: This chapter explains the fundamental concepts of network security, associated challenges, and existing solutions. It also emphasizes the application of machine learning and reinforcement learning techniques in detecting network-based threats.

Chapter III: This chapter presents a review of existing research on network attack detection using machine learning and reinforcement learning methods. It includes a detailed literature survey and identifies key research gaps.

Chapter IV: This chapter analysis of cybersecurity threat detection system including traditional, emerging trends, machine learning, deep learning based, hybrid ML-RL based techniques, evaluation parameters, experimental results, and summary.

Chapter V: This chapter focuses on the architecture of the proposed model based on machine learning algorithms, experimental setup, system configuration, dataset details, system design, experimental results and analysis, ablation study, and summary.

Chapter VI: This chapter describes the architecture of the proposed model, explaining the integration of reinforcement learning with hybrid optimization algorithms for feature selection and policy refinement. The process of training, learning, and agent-based decision optimization is discussed in detail. Experimental setup, dataset details, and performance results obtained from the proposed approach. The outcomes are compared with existing models to evaluate improvements in detection accuracy, computational efficiency, and response rate.

Chapter VII: This chapter illustrates the the development of a mechanism for tracking and blocking cybersecurity threats, system design and working mechanism, results, and analysis.

Chapter VIII: This chapter presents the experimental results along with a detailed evaluation of the proposed framework. It supports the research claim that the model enhances detection

accuracy, decreases response time, and improves adaptability for efficient cybersecurity threat detection.

Chapter IX: This chapter summarizes the overall research findings and contributions. It also discusses future research directions, including the use of reinforcement learning for more adaptive and real-time threat detection.

Publications on PhD work

International Journals:

- [1] Rakhi Kalantri and Rajesh Bansode, “Advancing Cyber Threat Detection with AI: Cutting-Edge Techniques and Future Trends “, *Journal of Information Systems Engineering and Management* , Vol. 10 No. 14s (2025), ISSN: 2468-4376 (Scopus Indexed). <https://jisem-journal.com/index.php/journal/article/view/2301>
- [2] Rakhi Kalantri and Rajesh Bansode, “Robust Framework for Detecting and Classifying Network-based Attacks using Ensemble Machine Learning Approach ,” *Journal of Applied Research and Technology*, (Accepted) (Unpaid Q3 Scopus Indexed Journals)
- [3] Rakhi Kalantri and Rajesh Bansode, “ Adaptive Multi-Agent Reinforcement Learning Framework for Cyber Attack Detection and Tracking,” *International Journal of Robust and Nonlinear Control* , 2026 . <https://onlinelibrary.wiley.com/doi/10.1002/rnc.70415> (Unpaid Q1 SCI and Scopus Indexed Journals)
- [4] Rakhi Kalantri and Rajesh Bansode, “Design and Development of Battle Royale Garra Rufa Fish Optimization with Dilated Residual Hierarchical Auto-Associative Polynomial CNN for Robust Cybersecurity Threat Detection,” submitted in *SN Computer Science* . (Q1 Unpaid Journal)

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- [1] Rakhi Kalantri and Rajesh Bansode, “ A Systematic Study on Cybersecurity Technologies Issues and Its Challenges,” *International conference on Intelligent Computing and Networking (IC-ICN2023)*, 24th -25th February 2023. <https://multiconw.co.in/wp-content/uploads/2023/05/IC-ICN-2023-Proceedings.pdf>
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